

# Cannabinoid Combination Targets *NOTCH1*-Mutated T-ALL Through the Integrated Stress Response Pathway

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## Abstract

In T-cell acute lymphoblastic leukemia (T-ALL) more than 50% of cases display autoactivation of Notch1 signaling, leading to oncogenic transformation. We have previously identified a specific chemovar of Cannabis that induces apoptosis by preventing Notch1 maturation in leukemia cells. Here, we isolated three cannabinoids from this chemovar that synergistically mimic the effects of the whole extract. Two were previously known, Cannabidiol (CBD) and Cannabidivarin (CBDV); whereas the third cannabinoid, which we termed 331-18A, was identified and fully characterized in this study. We demonstrated that these cannabinoids act through Cannabinoid receptor type 2 and TRPV1 to activate the integrated stress response pathway by depleting intracellular  $\text{Ca}^{2+}$ . This is followed by increased mRNA and protein expression of ATF4, CHOP and CHAC1, which is hindered by inhibiting the upstream initiation factor eIF2 $\alpha$ . The increased abundance of CHAC1 prevents Notch1 maturation, thereby reducing the levels of the active Notch1 intracellular domain, and consequently decreasing cell viability and increasing apoptosis. Treatment with the three isolated molecules resulted in reduced tumor size and weight in-vivo and slowed leukemia progression in mice models. Altogether, this study elucidated the mechanism of action of three distinct cannabinoids in modulating the Notch1 pathway, and constitutes an important step in the establishment of a new therapy for treating *NOTCH1*-mutated diseases and cancers such as T-ALL.

### eLife assessment

This **important** study follows up on previous work defining the anti-leukemic effects of a previously characterized cannabis extract on Notch-activated T cells and identifies several pathways that mediate its anti-cancer activity including the ER calcium and integrated stress response. The evidence is **solid**, but several concerns remain including the over reliance on a single cell line for the majority of the studies and lack of integration of the observations with existing literature

## Introduction

T-cell acute lymphoblastic leukemia (T-ALL) is an aggressive cancer, characterized by immature T-lymphoblasts (1, 2). In over 50% of cases, activating mutations in *NOTCH1* were identified, generating much interest in targeting the Notch1 signaling pathway in this disease (3–5). The Notch1 protein is a single-pass transmembrane receptor that is expressed on the plasma membrane. It regulates evolutionarily conserved signaling that controls developmental processes, cell fate determination and tissue homeostasis. It is a member of the Notch receptor family, which consists of four receptors (Notch1–4) in humans (4, 6). The Notch1 receptor-signaling pathway is activated through a series of proteolytic cleavages at three different sites (S1–S3). S1 cleavage is carried out by a Furin-like convertase in the trans-Golgi apparatus, resulting in a heterodimeric receptor that is transported to the membrane where it can bind ligands from a neighboring cell. Upon binding of a ligand, cleavage is performed at the next site and the activation of the Notch1 receptor-signaling pathway is initiated, enabling the cleavage of S3 in the transmembrane region that causes the release of the Notch1 intracellular domain (NICD), which translocates to the nucleus where it promotes transcription of target genes involved in cell growth (7).

Aberrant Notch1 activity is associated with many other leukemia types and other cancers, including prostate, lung, ovarian, and renal (8–12). A new and promising emerging field in cancer treatment is medical Cannabis and its unique active compounds, the phytocannabinoids. Phytocannabinoids affect the body by means of the endocannabinoid system (eCBS), a ubiquitous neuromodulatory signaling system that has widespread functions throughout the body. Notable eCBS receptors include cannabinoid receptor type 1 (CB1) and CB2, which belong to the family of G protein-coupled receptors, and transient receptor potential vanilloid type 1 (TRPV1), which is an ion channel (13). The eCBS participates in many physiological activities, including the regulation of  $\text{Ca}^{2+}$  homeostasis (14–16).

Medical Cannabis is already commonly used by cancer patients for its palliative effects, it was shown to stimulate appetite while alleviating symptoms such as nausea, vomiting and pain (17). However, accumulating evidence demonstrated phytocannabinoids directly affect tumor development in cell lines and animal models by modulating key cell-signaling pathways (18–20). Studies demonstrated that purified phytocannabinoids can inhibit proliferation, metastasis, and angiogenesis and exert pro-apoptotic effects in a variety of cancer cell types such as lung, breast, prostate, skin, intestine, glioma, lymphoma, pancreas and uterine cancers (21–23). In preclinical models, treatment with phytocannabinoids led to tumor regression of different cancer types (24, 25).

Concerning T-ALL, some studies using phytocannabinoids and their respective formulations (such as Dronabinol) have shown anti-proliferative as well as proapoptotic effects in selected leukemia cell lines and native leukemia blasts cultured *ex-vivo* (26–28). Nevertheless, the mechanisms of phytocannabinoid-mediated antitumor effects are not yet fully understood. Most of the studies that examined the anti-cancer effects of Cannabis have focused mainly on the two major phytocannabinoids (–)-*trans*- $\Delta^9$ -tetrahydrocannabinol ( $\Delta^9$ -THC) and Cannabidiol (CBD), but these are just two of more than 140 different phytocannabinoids that have been identified in different Cannabis chemovars (29, 30).

We have previously demonstrated that different Cannabis extracts with unique phytocannabinoid compositions impaired the survival and proliferation of specific cancer cell lines, suggesting that the effect of a particular Cannabis extract on a specific cancer cell line relies on its chemical composition (31). Following this framework, we successfully matched a specific CBD-rich Cannabis extract to T-ALL leukemia cells that harbor a *NOTCH1* mutation (32). In the present study, we investigated which specific metabolites in the whole extract are responsible for T-ALL elimination, and elucidated the molecular mechanism by which they mediate their antitumor effects in this type of cancer.

## Results

We have previously shown a specific CBD-rich Cannabis extract selectively induces apoptosis in T-ALL leukemia cell lines that harbor a *NOTCH1* mutation. Treatment led to accumulation of the full-length immature form of Notch1 in the membrane, correspondingly with reduced NICD protein expression and transcription activity (32).

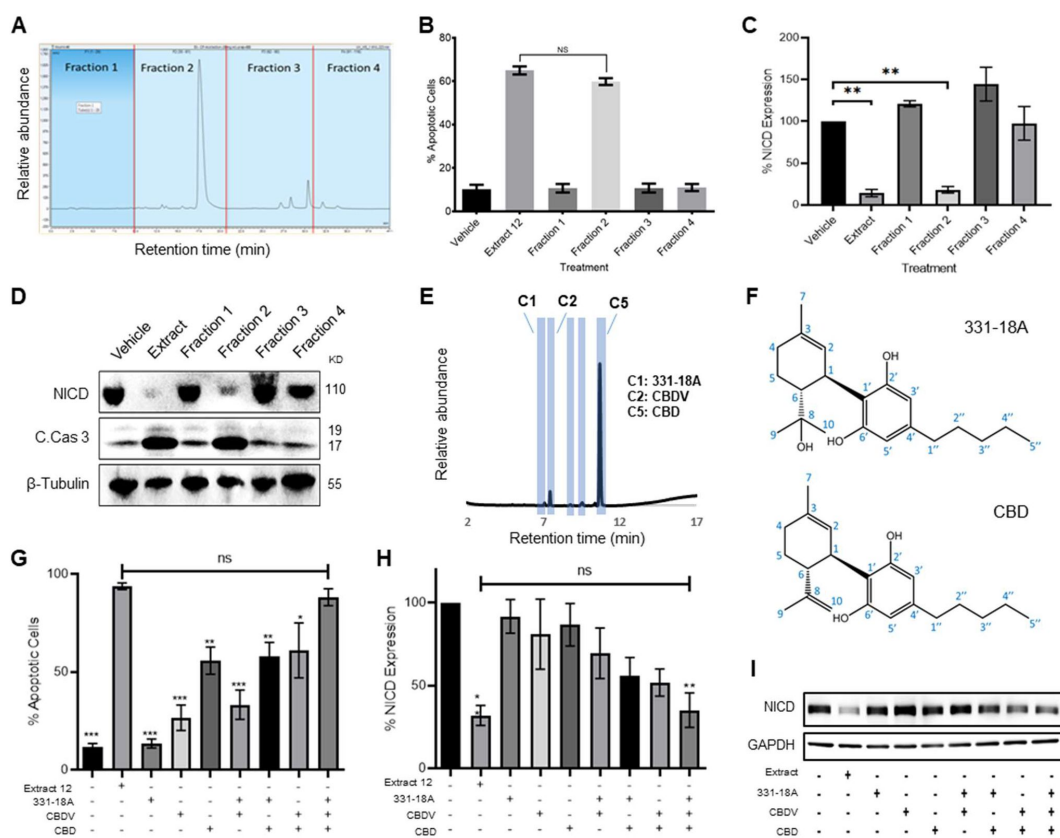
### A combination of three phytocannabinoids induces apoptosis and reduces NICD expression

The CBD-rich extract was more potent than pure CBD or other high-CBD chemovars (32). This implied a specific combination of metabolites from the plant potentiate the full antitumor effect. Using a semi-preparative HPLC, we separated the whole extract into four 10-minute retention time fractions (Fig. 1A). A full list of the phytocannabinoids constituting each fraction is presented in supplementary table S1. We assessed the effect of each fraction on the viability of MOLT-4 cells (fig. S1A) and found fraction 2 reduced viability to a similar extent as the whole extract. Further analysis revealed that out of the four fractions, only fraction 2 induced apoptotic cell death (Fig. 1B), reduced the protein levels of the NICD (Fig. 1C, D) and elevated the levels of cleaved caspase-3 (Fig. 1D). Thus, we excluded all the phytocannabinoids present in the other three fractions, such as  $\Delta^9$ -THC in fraction 3.

To pinpoint the active phytocannabinoids in fraction 2, we further separated this fraction into individual phytocannabinoids using centrifugal partition chromatography (CPC) followed by semi-preparative HPLC (C1-C5 in Fig. 1E). We identified the specific compounds in the isolated peaks (Supplementary table S2), C2 was identified as cannabidivarin (CBDV) and C5 as CBD. C3 and C4 were identified as Cannabidiolic acid (CBDA), the precursor to CBD, and as CBD-C4, respectively; but their concentrations were very low. C1 was identified as 331-18A, a phytocannabinoid first identified by our team in decarboxylated CBD-rich Cannabis chemovars (29), we characterized 331-18A using ESI-LC/MS/MS as having a deprotonated  $m/z$  of 331.2279 and MS/MS fragments resembling those of CBD with the addition of one hydroxyl group. To confirm the identification, the purity of the three isolated compounds was determined by UHPLC/UV (fig. S2A-C).

Here, we further elucidated the chemical structure of 331-18A by nuclear magnetic resonance (NMR) with CBD as the reference (Supplementary Fig. S3 and table S3). The NMR data of CBD was in close agreement with the literature (33). The spectra of CBD and 331-18A were very similar with the exception of the disappearance of the external double bond (C8-C10) and the appearance of a tertiary alcohol instead. The absolute stereochemistry of 331-18A was confirmed by specific rotation determination with a polarimeter in which 331-18A showed  $[\alpha]^{20}_D -56.85$  which is comparable with the  $[\alpha]^{20}_D -64.16$  of both the CBD that was isolated from the fraction (C5) and a commercial standard of CBD. These results suggested that similarly to CBD, the two substituents at positions 1 and 6 of the cyclohexene ring are in a *trans* configuration (Fig. 1F).

To examine the cytotoxic properties of the isolated compounds and their combinations on MOLT-4 cells, the isolated phytocannabinoids were prepared in the same amounts and proportion as observed in the whole extract; for example, if 1  $\mu$ g of the whole extract contained 50% CBD, we used 0.5  $\mu$ g of the purified CBD and 1  $\mu$ g of the whole extract in the respective treatments. Compared to the whole extract, the phytocannabinoids 331-18A and CBDV alone did not induce apoptosis or had minor effects on the cells, and CBD alone induced only 55% cell death compared to the whole extract (Fig. 1G). When CBD was combined with either 331-18A or CBDV, the percent of apoptotic cells increased to 58% and 61%, respectively. The combination of all three phytocannabinoids led to the greatest cytotoxic effect and only this combination was comparable to the whole extract ( $88.29\% \pm 4.31\%$  and  $93.85\% \pm 1.64\%$ , respectively). We further evaluated the effect of the three phytocannabinoids separately or in different combinations on the protein



**Fig. 1.**

### Identifying specific phytocannabinoids responsible for the cytotoxic effect of the whole extract.

(A) A semi-preparative HPLC-UV chromatogram of the whole extract showing the four fractions collected every 10 minutes. (B-D) MOLT-4 cells were treated with either vehicle, whole extract or fractions 1-4 (3 µg/mL) and (B) apoptosis was analyzed after 24 h (N=3) via Annexin V/PI, (C-D) NICD and cleaved caspase-3 (C. Cas. 3) expressions were evaluated (N=3) after 3 h with β-tubulin as the loading control. A representative blot is shown. (E) UHPLC/UV chromatogram of fraction 2, the specific compounds constituting the fraction are marked (C1-C5). (F) Chemical structure and peak assignment of CBD and 331-18A according to <sup>1</sup>H and <sup>13</sup>C NMR. Atom numbering is according to the monoterpene numbering system. (G-I) MOLT-4 cells were treated with the whole extract (3 µg/mL) or with 0.06 µg/mL 331-18A, 0.06 µg/mL CBDV and 1.5 µg/mL CBD, their corresponding concentrations in the extract, and their different combinations. Cells were assessed for (G) apoptosis after 24 h (N=3) via Annexin V/PI assay and (H-I) NICD expression after 3 h (N=3) with GAPDH as the loading control. Results are presents as mean ± SEM and statistically analyzed with One-way ANOVA (\*p<0.05, \*\*p<0.01, \*\*\*p<0.001).

expression of NICD (**Fig. 1H-I**) and found that all combinations of phytocannabinoids led to a significant reduction in NICD expression, but only when all the three were combined, NICD levels decreased to the same extent as with the whole extract.

### 331-18A, CBDV and CBD mediate their effect

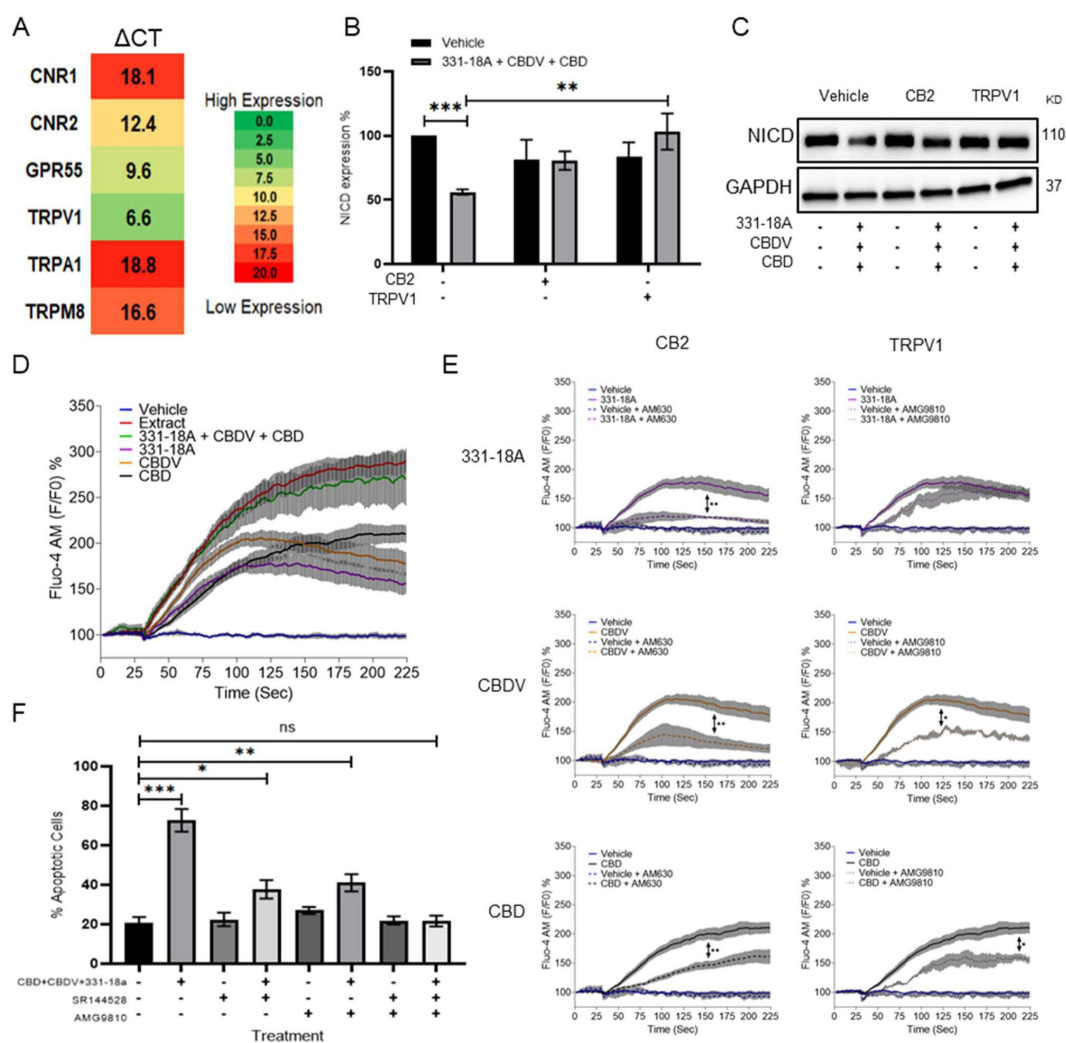
#### through CB2 and TRPV1 to trigger $\text{Ca}^{2+}$ flux

Phytocannabinoids usually operate through specific cannabinoid receptors in the eCBS. To elucidate which endocannabinoid receptors mediate the antitumor effect, we first evaluated in MOLT-4 cells the mRNA expression of *CNR1* and *CNR2* (for CB1 and CB2, respectively), as well as G protein-coupled receptor 55 (*GPR55*), *TRPV1*, Transient receptor potential cation channel, subfamily A, member 1 (*TRPA1*) and Transient receptor potential cation channel subfamily M (melastatin) member 8 (*TRPM8*), (**Fig. 2A**). The  $\Delta\text{CT}$  results showed low expression for *CNR1*, *TRPA1* and *TRPM8*, medium expression for *CNR2* and *GPR55*, and high expression for *TRPV1*. We tested antagonists for those receptors that had either high or medium expression: *TRPV1* (AMG9810), CB2 (AM630 or SR-144,528), *GPR55* (CID), and general Gq GPCR (BIM-46187) as a control, and evaluated whether they were able to rescue NICD expression upon treatment with the whole extract (fig. S4A-B). Only the CB2 and *TRPV1* antagonists rescued the expression of NICD upon treatment with the whole extract, therefore we tested whether they were able to rescue NICD expression after treatment with the combination of 331-18A, CBDV and CBD (**Fig. 2B-C**). The CB2 antagonist rescued the expression from 56% to 87%, and the *TRPV1* antagonist rescued NICD expression to 100% by itself.

*TRPV1* is a  $\text{Ca}^{2+}$  channel and eCBS receptors are known to trigger  $\text{Ca}^{2+}$  flux (15, 16). Therefore, as the next step we tested the effect of the three cannabinoids on cytosolic  $\text{Ca}^{2+}$  levels. We used the Fluo-4 calcium probe to measure the effect of 331-18A, CBDV and CBD combination on MOLT-4. Each of the molecules separately caused some elevation in  $\text{Ca}^{2+}$ , but only their combination had an effect on  $\text{Ca}^{2+}$  to the same extent as the whole extract (**Fig. 2D**). We tested  $\text{Ca}^{2+}$  levels when cells were pretreated with the antagonists to CB2 and *TRPV1*. Both antagonists inhibited the elevation in response to the whole extract (Supplementary fig. S3C, D). When 331-18A, CBDV and CBD were tested separately with antagonists to either CB2 or *TRPV1* (**Fig. 2E**), CB2 but not *TRPV1* antagonist diminished 331-18A induced  $\text{Ca}^{2+}$  increase, indicating this cannabinoid exerts its effect through CB2, while both antagonists significantly decreased the  $\text{Ca}^{2+}$  elevation by CBDV and CBD. Next, we assessed whether inhibition of the receptors rescues MOLT-4 cells from the cytotoxic properties of the cannabinoid combination (**Fig. 2F**). Treatment with the inhibitors by themselves, separately or both together, did not induce MOLT-4 cell death. Upon treatment with the combination of 331-18A, CBDV and CBD, inhibition of CB2 only partially rescued the cells from treatment-induced apoptosis, reducing the apoptotic rate from 75% of the combination-treated cells to 40%. However, the apoptotic rate was still significantly higher than that of the untreated control. Inhibition of *TRPV1* only also partially rescued the cells, to ~50%, but again the apoptotic rate remained significantly increased relative to the untreated control. Inhibition of both CB2 and *TRPV1* simultaneously rescued the cells from treatment-induced apoptosis, confirming the effect of the three cannabinoids is mediated through both receptors.

### Notch1 downregulation by cannabinoids is mediated through the integrated stress response pathway

To elucidate the mechanism by which the Notch1 pathway is down-regulated following treatment, we used RNA-seq analysis to measure and compare the gene expression profile of MOLT-4 cells after 3 hours of treatment with either vehicle or the whole extract (**Fig. 3A**). A list of the ten most increased- and decreased-abundance genes following treatment is presented in supplementary table S4. ChaC glutathione-specific  $\gamma$ -glutamylcyclotransferase 1 (*CHAC1*), a negative regulator of the Notch signaling pathway (34), was identified as the second most increased-abundance gene. *CHAC1* acts by inhibiting the furin-like cleavage of Notch1, preventing



**Fig. 2.**

### 331-18A, CBDV and CBD mediate their effect through CB2 and TRPV1.

(A) The mRNA levels of receptors *CNR1* (CB1), *CNR2* (CB2), *GPR55*, *TRPV1*, *TRPA1* and *TRPM8* were evaluated by qRT-PCR. Gene expression levels were calculated as  $\Delta$ CT normalized to *GUSB* housekeeping gene. Results are presented as mean expression of three biological replicates. High  $\Delta$ CT values indicate low receptor expression. (B) MOLT-4 cells were pretreated with 50  $\mu$ M antagonist to CB2 (AM630) or TRPV1 (AMG9810) for 30 min, then treated with the combination of the three cannabinoids for 3 h. NICD expression was evaluated by western blot (N=3) with GAPDH as the loading control and statistically analyzed with an unpaired student's t-test (\*\* $p$ <0.01, \*\*\* $p$ <0.001). (C) Representative blots of B. (D) Calcium release by MOLT-4 cells was measured with the Fluo-4 calcium probe immediately following treatment with either vehicle, whole extract, a combination of 331-18A, CBDV and CBD, and each cannabinoid separately. The calcium curves represent an average of three independent experiments. (E) Reduction of each phytocannabinoid-induced calcium release by pretreatment with 50  $\mu$ M of antagonist to CB2 (AM630) or TRPV1 (AMG9810) for 30 min. The calcium curve for each compound is given again in this graph to compare the effects with the antagonists (N=3, two-way ANOVA, \* $p$ <0.05, \*\* $p$ <0.01). (F) MOLT-4 cells were pretreated with 50  $\mu$ M antagonist to CB2 (SR-144,528) or TRPV1 (AMG9810) or both for 30 min, then treated with the combination of the three cannabinoids or left untreated. Apoptosis was analyzed after 24 h (N=3) by Annexin V/PI. Results are presented as mean  $\pm$  SEM and statistically analyzed with one-way ANOVA (\* $p$ <0.05, \*\* $p$ <0.01, \*\*\* $p$ <0.001).

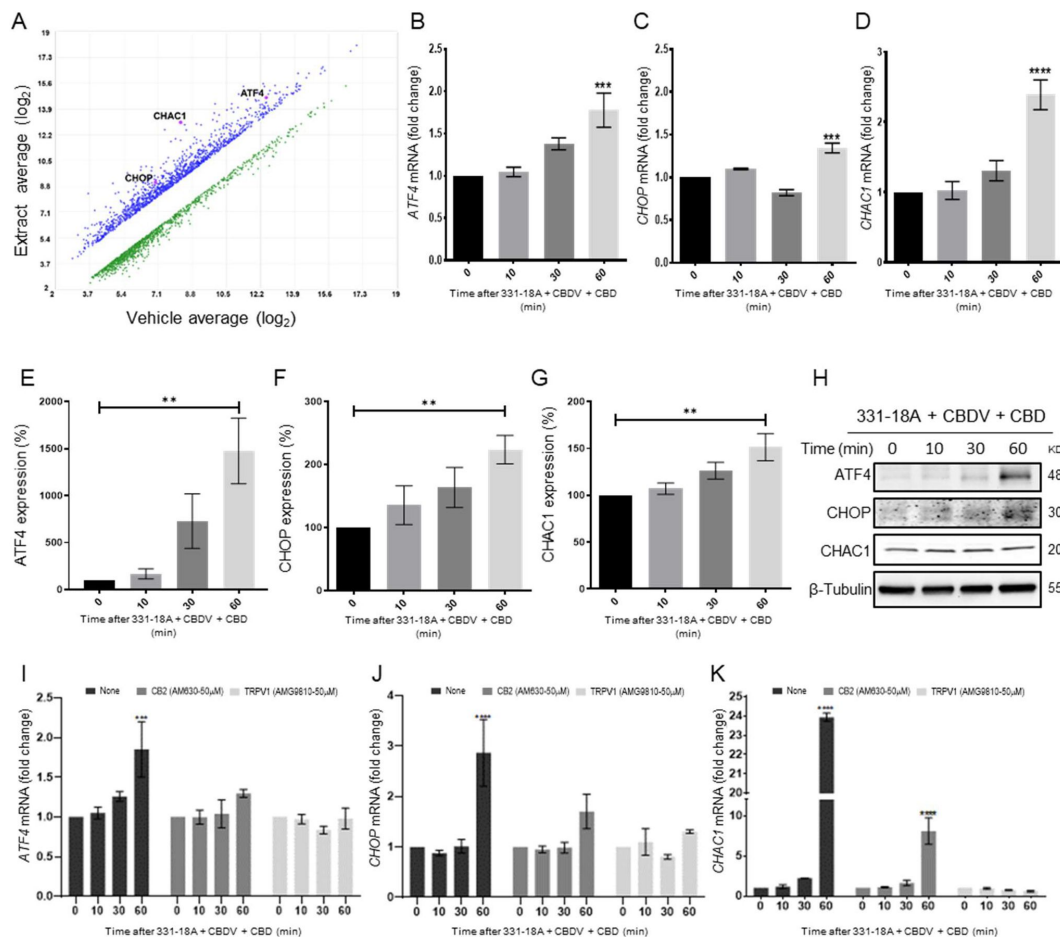


the formation of the mature form of Notch1 (35), as we have previously demonstrated upon treatment with the whole extract (32). C/EBP homologous protein (*CHOP*) and Activating transcription factor 4 (*ATF4*), ER-stress signaling pathway markers that are upstream to *CHAC1* (36), had also a significant increase in their abundance. We confirmed the RNA-seq results by qRT-PCR showing that the mRNA of *ATF4*, *CHOP* and *CHAC1* is indeed significantly increased after treatment with a combination of 331-18A, CBDV and CBD (Fig. 3B-D, respectively) or the whole extract (fig. S3E-G). A significant increase was apparent 60 min after treatment. Using specific antibodies to *ATF4*, *CHOP* and *CHAC1*, we verified the protein expression level of all three was also significantly increased at 60 min after treatment (Fig. 3E-H). The protein expression levels of *CHAC1* in MOLT-4 cells were also significantly increased upon treatment with the whole extract (fig. S3H, I).

To further understand the mechanism, we evaluated how pretreatment with antagonists to the cannabinoid receptors through which 331-18A, CBDV and CBD exert their effect, CB2 and TRPV1, affected *ATF4*, *CHOP* and *CHAC1* expression following treatment (Fig. 3I-K). Inhibition of CB2 reduced the treatment-induced increase in mRNA expression, but a trend of increase for *ATF4* and *CHOP* remained at 60 minutes, and *CHAC1* expression was significantly increased. Inhibition of TRPV1 rescued from treatment-induced induction of all three targets. Similar effects were found when pretreatment with antagonists to CB2 and TRPV1 was followed by treatment with the whole extract (fig. S3J-L); and the increase in the expression levels of *CHAC1* protein was abolished when MOLT-4 cells were treated by either antagonist (fig. S3M). These results demonstrate that CB2 and TRPV1 participate in inducing *CHAC1* expression and preventing Notch1 maturation, resulting in a decrease in NICD release and cell death.

We therefore speculated that the combination of the three cannabinoids caused the activation of *ATF4*-*CHOP*-*CHAC1* signaling by  $\text{Ca}^{2+}$  depletion and ER-stress (37). To further verify the role of *ATF4*-*CHOP*-*CHAC1* signaling in mediating the downregulation of the Notch1 pathway, we inhibited the activity of the upstream target eukaryotic Initiation Factor-2 $\alpha$  (eIF2 $\alpha$ ) in MOLT-4 cells using integrated stress response inhibitor (ISRIB), which inhibits the activity of eIF2 $\alpha$  without affecting its phosphorylation (38) (fig. S5A-B). Then, we treated the cells with either vehicle or the combination of the cannabinoids. After 24 hours, blocking the activity of eIF2 $\alpha$  prevented the cytotoxic effect of 331-18A, CBDV and CBD combination (Fig. 4A) or of the whole extract (fig. S5C). Moreover, eIF2 $\alpha$  inhibition prevented the reduction in NICD expression upon treatment with either the whole extract (fig. S5D-E) or the cannabinoid combination (Fig. 4B-C). Inhibition of eIF2 $\alpha$  also rescued the cells from the induced increase in the mRNA expression of *ATF4*, *CHOP* and *CHAC1* upon treatment with the combination of the cannabinoids (Fig. 4D-F). Thus, inhibition of the integrated stress response pathway, eIF2 $\alpha$ -*ATF4*-*CHOP*-*CHAC1* signaling, rescued MOLT-4 cells from the cytotoxic effect of the combination of the three cannabinoids and prevented the reduction in NICD expression, as well as the increased transcription of *ATF4*, *CHOP* and *CHAC1* upon treatment.

To further investigate the interaction between 331-18A, CBDV and CBD, we assessed their effect on the viability of MOLT-4 cells in different relative proportions. First we tested the ability of each separate cannabinoid to inhibit the proliferation of MOLT-4 cells in doses ranging 0-2  $\mu\text{g}/\text{ml}$  (Fig. 5A), which is below and above the overall concentration for which we found an effect for the cannabinoid combination. 331-18A and CBDV affected viability from a dose of 1  $\mu\text{g}/\text{ml}$  and CBD from a dose of 0.5  $\mu\text{g}/\text{ml}$ . Next, we calculated the combination index of every pair and all three cannabinoids in concentrations ranging from 0-2  $\mu\text{g}/\text{mL}$  (Fig. 5B-E). The most synergistic area is displayed in red. We calculated the total synergy score for the combination of all three cannabinoids relative to the pairwise combinations (Fig. 5F). According to Highest Single Agent (HSA) reference model (39), a combination of CBDV with either 331-18A or CBD results in a likely antagonistic interaction (score < -10), a combination of 331-18A and CBD results in a likely additive interaction (score from -10 to 10), and only the combination of all three cannabinoids is likely synergistic (score > 10, Fig. 5G).

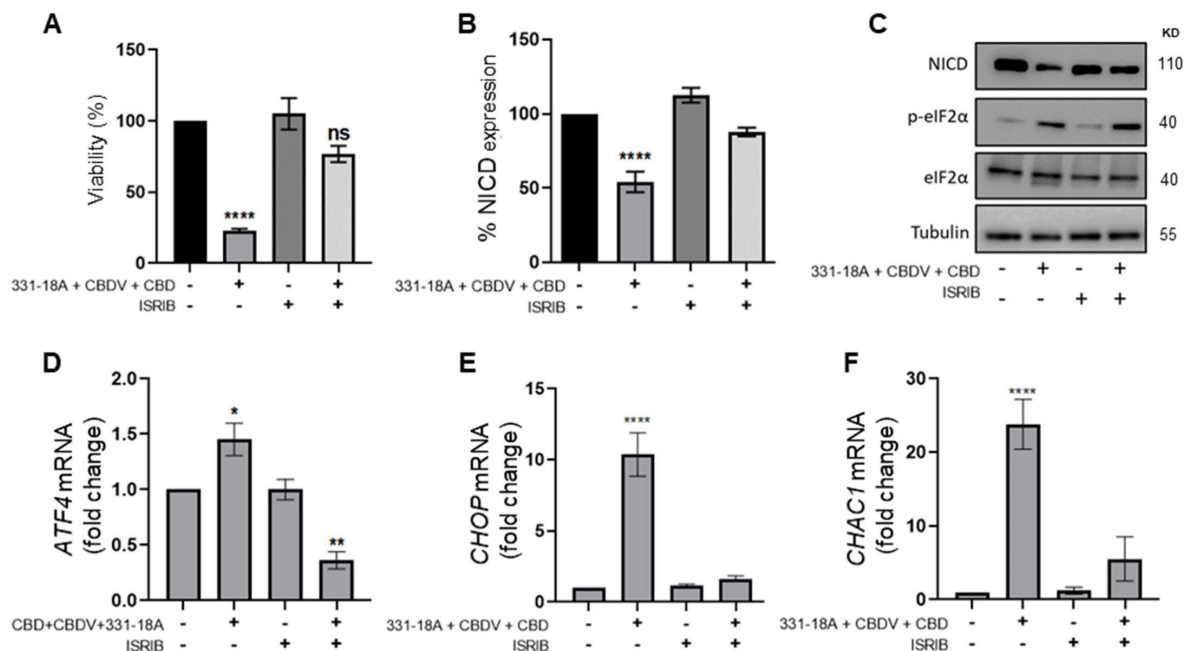


**Fig. 3.**

### Notch1 downregulation by 331-18A, CBDV and CBD combination is mediated via ATF4-CHOP-CHAC1 signaling pathway.

(A) MOLT-4 cells were treated for 3 h with either vehicle or the whole Extract (3  $\mu\text{g}/\text{mL}$ ) and an RNA-seq (affymetrix) scatter plot presents the differential expression; the increased abundance genes *CHAC1*, *CHOP* and *ATF4* are marked (N=3). (B-D) MOLT-4 cells were treated for 3 h with 331-18A, CBDV and CBD combination (N=3), the mRNA expression of *ATF4*, *CHOP* and *CHAC1* genes was evaluated with qRT-PCR at different time points (0-60 min). Differences are presented as fold-change  $\pm$  SEM and statistically analyzed with one-way ANOVA (\*\*\* $p$ <0.001, \*\*\*\* $p$ <0.0001). (E-G) MOLT-4 cells were treated as in B-D and the protein expression of ATF4, CHOP and CHAC1 were evaluated with  $\beta$ -tubulin as the loading control. Intensity analysis was performed on three independent experiments (N=3) and statistically analyzed with an unpaired Student's *t*-test (\*\* $p$ <0.01). (H) Representative blots. (I-K) Analysis of ATF4, CHOP and CHAC1 mRNA expression at different time points following pretreatment with CB2 and TRPV1 antagonists for 30 min and then treatment with the cannabinoid combination (N=3), analyzed with two-way ANOVA (\*\*\* $p$ <0.001, \*\*\*\* $p$ <0.0001).

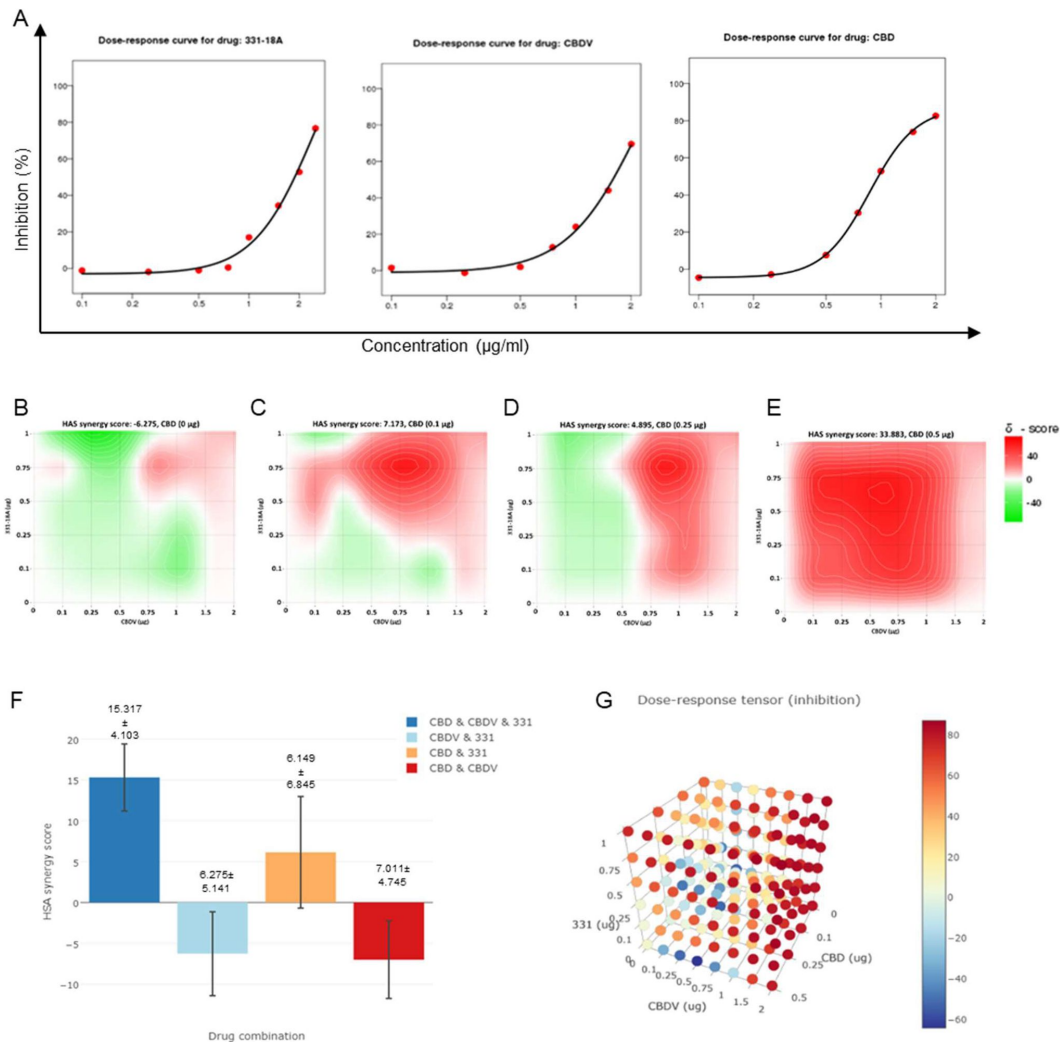




**Fig. 4.**

#### 331-18A, CBDV and CBD combination activates eIF2 $\alpha$ .

MOLT-4 cells were pretreated for 30 min with the eIF2 $\alpha$  inhibitor ISRIB (150  $\mu$ M) or left untreated, then treated with vehicle or the combination of 331-18A, CBDV and CBD for 3 h. **(A)** Viability after 24 hrs was assessed with XTT (N=3). **(B)** NICD protein expression was assessed (N=3) with  $\beta$ -Tubulin as the loading control (unpaired Student's *t*-test, \**p*<0.05). **(C)** A representative image also showing the protein levels of phosphorylated eIF2 $\alpha$  (Ser<sup>51</sup>) and total eIF2 $\alpha$ . **(D-F)** *ATF4*, *CHOP* and *CHAC1* gene expression was assessed via qRT-PCR 4 hrs after treatment (N=3). Results are presented as mean  $\pm$  SEM and statistically analyzed with one-way ANOVA (\*\*\*\**p*<0.0001).



**Fig. 5.**

### All three cannabinoids must be combined for a synergistic effect.

(A) MOLT-4 cell death (N=3) was assessed by XTT following treatments with concentrations ranging 0-2 µg/ml and a dose-response curve for each cannabinoid separately was plotted. (B-E) Synergy distribution for different ratios of cannabinoids was calculated by HSA model with SynergyFinder web application (v 2.0), representative dose-response matrices are presented for CBD at (B) 0, (C) 0.1, (D) 0.25 and (E) 0.5 µg/ml. (F-G) Best synergy scores according to HSA model presented for the triple combination relative to the two-cannabinoid combinations.

## Inhibition of cancer progression in-vivo by 331-18A, CBDV and CBD combination

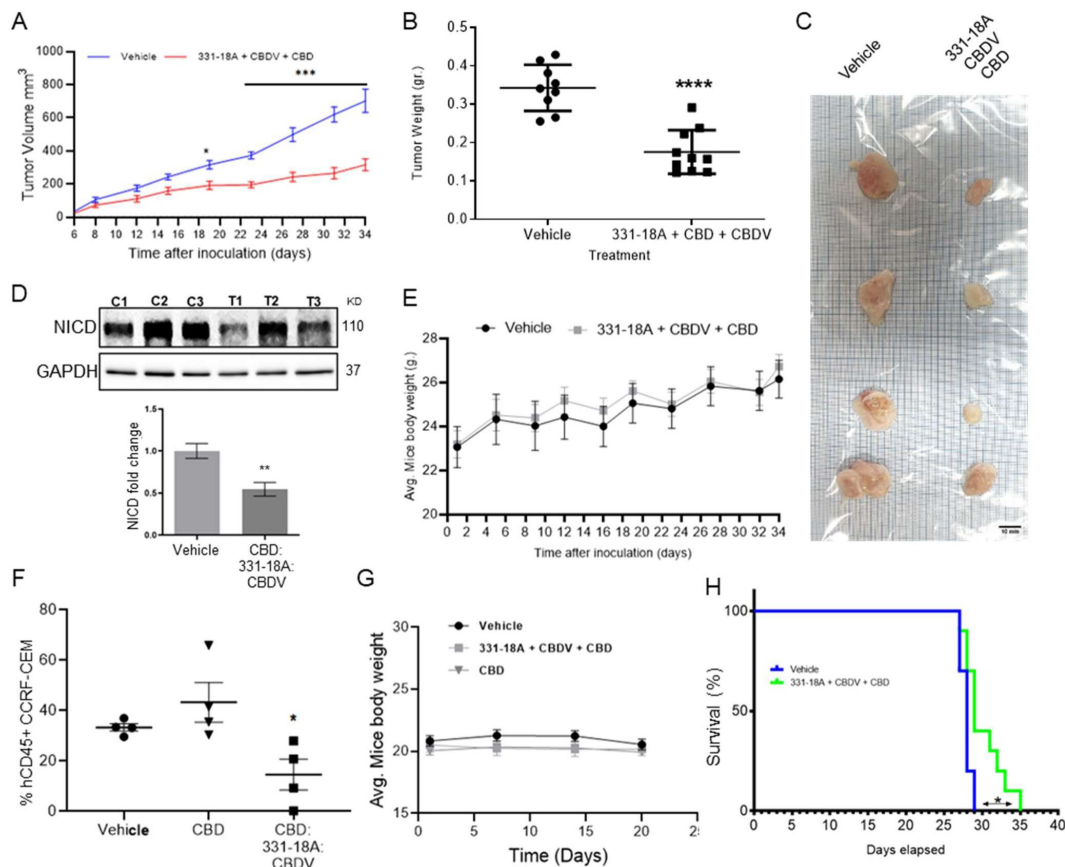
We further evaluated the anti-tumor properties of the three cannabinoids using several mouse models. The first model was a subcutaneous tumor model for which we have already demonstrated the effectivity of the whole extract (32). We engrafted subcutaneously Nonobese diabetic-severe combined immunodeficiency (NOD/Scid) mice with MOLT-4 cells, then treated with vehicle only or the combination of the cannabinoids from when tumors reached a volume of 100 mm<sup>3</sup> (day 7), on alternating days, for a duration of four weeks. Treatment with a combination of 331-18A, CBDV and CBD significantly inhibited tumor growth (Fig. 6A-C). Significant differences were apparent approximately 3 weeks after treatment started (Fig. 6A), and at the endpoint, 34 days from the injection of cells, the average weight of tumors was significantly lower in the treated group compared to the control group (Fig. 6B). Excised tumors (Fig. 6C) were analyzed for their cannabinoid content according to a method recently developed by our group for biological tissues (40). Using ESI-LC/MS, we identified and quantified each of the three cannabinoids in the tumors [16.0 ± 6.7, 22.3 ± 10.4 and 265.8 ± 109.7 ng/g for CBDV, 331-18A and CBD, respectively (average ± S.E.M., N=3)]. Notably, the ratio of the cannabinoids analyzed in the tumors was similar to the ratio given to the mice in the treatment protocol. In addition, a significant decrease was observed in NICD expression in the tumors treated with the cannabinoids (Fig. 6D). To assess possible toxicity, the body weight of the mice was monitored throughout the experiment (Fig. 6E). There were no significant differences in the average weight of mice in the control group and the treatment group.

Next, we assessed the cannabinoids using an intravenous model which better resembles leukemogenesis (41, 42). We engrafted NOD-SCID IL2ry-null (NSG) mice that are a better host for human immune cells with CCRF-CEM, Notch1 mutated T-ALL cells. The mice were treated with a combination of the three cannabinoids, and to verify the need for the combination, one group was also treated with pure CBD in the same overall dose. At the endpoint of the experiment, the amount of human CD45+ cells detected in the bone marrow (BM) was significantly decreased in the mice treated with the combination of the three cannabinoids compared to the control, and compared to pure CBD (Fig. 6F). In this model as well, there were no significant differences in the average weight of the mice (Fig. 6G). The combination of 331-18A, CBDV and CBD was as effective as the whole extract (fig. S6A-B). In addition, the administration of the cannabinoid combination significantly prolonged survival in treated mice compared to the survival of vehicle-treated control mice (Fig. 6H), to a similar degree to the whole extract (fig. S6C).

To examine the clinical relevance, we utilized a patient-derived xenografts (PDX) model with primary cells from a T-ALL patient. After allowing 35 days from transplantation for leukemic cell establishment, mice were treated for three weeks with either vehicle or whole extract containing the three cannabinoids. In mice that were treated with the extract, we found a significant reduction in the burden of leukemia in the bone marrow (fig. S7A), peripheral blood (fig. S7B) and spleen (fig. S7C). The weight of the spleen was also significantly lower in extract-treated mice (fig. S7D, E). Treatment with the whole extract did not affect the weight of the mice.

## Discussion

Targeting Notch signaling components has generated much interest for its therapeutic potential. However, so far efforts to develop treatments that target Notch signaling have been unsuccessful (43). Cannabis, in its many forms, is already being prescribed to cancer patients all over the world, mainly as a palliative treatment for the inhibition of nausea and emesis associated with chemotherapy, appetite stimulation, pain relief and relief from insomnia (17). On top of that, recent studies have demonstrated that phytocannabinoids also possess anti-cancer potential (24, 25, 31, 44–46).



**Fig. 6.**

### Inhibition of cancer progression in-vivo by a combination of 331-18A, CBDV and CBD.

(A) Female NOD/Scid mice (N=4-5/group, two independent experiments) were engrafted subcutaneously with  $1 \times 10^6$  MOLT-4 cells. After seven days, the mice were randomly divided into two groups and alternate-day treated intraperitoneally with either vehicle or a combination of 331-18A (1.11 mg/kg), CBDV (1.11 mg/kg) and CBD (17.77 mg/kg). Ectopic tumor volume was measured using a vernier caliper and calculated according to the formula  $(\text{length} \times \text{width}^2) \times 0.5$ . The average difference between groups was statistically analyzed by Bonferroni's multiple comparisons test (\* $p < 0.05$ , \*\*\* $p < 0.001$ ). (B) After 34 days of treatment, the mice were sacrificed and tumors were excised and weighed (N=9). The difference in the weights of the tumors between treatment groups was statistically analyzed by unpaired Student's *t*-test (\*\*\*\* $p < 0.0001$ ). (C) Representative photograph of excised tumors. (D) A representative blot (top) and fold change (bottom) of NICD protein expression in excised tumors at day 34 after treatment with either vehicle (C) or a combination of the three cannabinoids (T), with GAPDH as the loading control. Data are presented as mean  $\pm$  SEM (N=3) and statistically analyzed by unpaired Student's *t*-test (\*\* $p < 0.01$ ). (E) Body weight (grams) in vehicle- and cannabinoids-treated mice. (F) Female NSG mice were intravenously injected with  $1 \times 10^6$  human CCRF-CEM cells. Mice were randomly divided into three groups (N=5/group) and treated intraperitoneally daily for five days, then left untreated for two days before starting another cycle, for a period of three weeks. Treatment groups were vehicle, pure CBD (30 mg/kg), and a combination of CBD (26.67 mg/kg), CBDV (1.67 mg/kg) and 331-18A (1.67 mg/kg) at the same overall concentration as pure CBD. After 24 days, the percentage of hCD45+ cells in the bone marrow was measured with flow cytometry and statistically analyzed by unpaired student's *t*-test (\* $p < 0.05$ ). (G) Body weight (grams) in vehicle- and cannabinoids-treated mice. (H) Survival analyses after CCRF-CEM injection followed by treatment with either vehicle or the cannabinoid combination (N=10/group), statistical differences were calculated with the Log-rank (Mantel-Cox) test (\* $p < 0.05$ ).

In the current study, we identified three unique phytocannabinoids that together selectively induce apoptosis in Notch1 mutated leukemia cells: CBD, CBDV and the novel phytocannabinoid 331-18A. CBD and CBDV are well-known and commercially available. However, to the best of our knowledge, the third phytocannabinoid 331-18A, denoted in the literature as 6,12-dihydro-6-hydroxy-cannabidiol (47), has not been previously identified in natural and/or decarboxylated Cannabis plants. We suggest that the need for all three cannabinoids for the optimal anti-tumor effect demonstrates the critical importance of elucidating the individual and polypharmacological effects of the entire molecular spectrum of Cannabis.

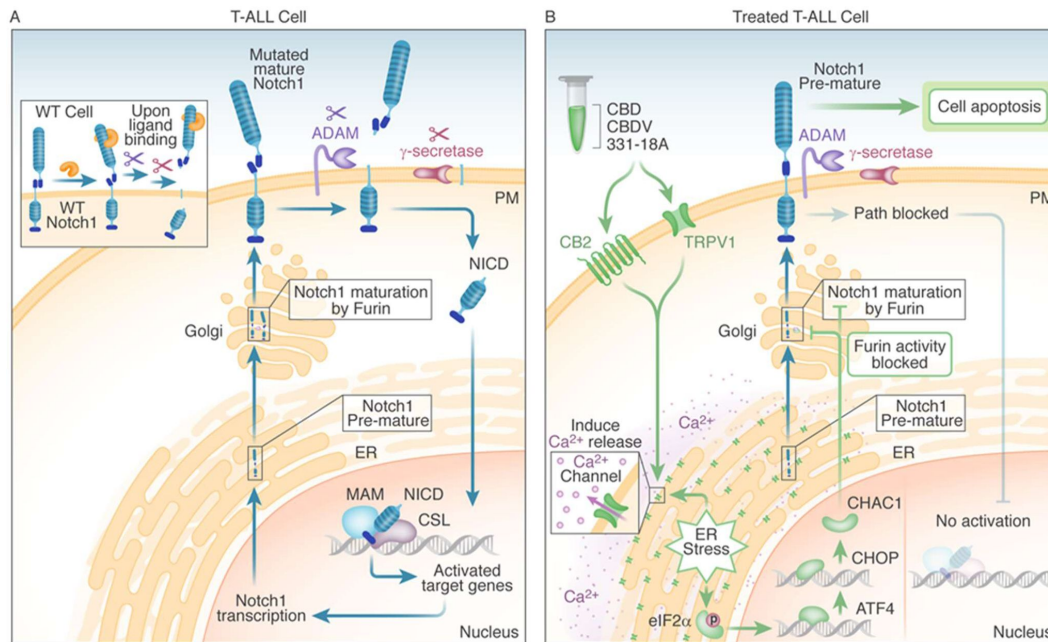
Although CBD by itself demonstrated some capacity to induce cell death, it was only when all three cannabinoids were combined that the same extent of response was achieved as treating with the whole extract. The Cannabis plant produces a vast array of over 140 phytocannabinoids and hundreds of other components, but the full extent of their biological effects and interactions is still not completely understood. Nevertheless, mounting evidence suggests that their combined effect as a whole is greater than that of each separate compound (48, 49). Recently, Blasco-Benito *et al.* (50), have shown that a  $\Delta^9$ -THC-rich extract was much more potent than pure  $\Delta^9$ -THC in producing antitumor responses in both cell culture and animal models of breast cancer. However, they were unable to identify the combinations of compounds that were responsible for this increased potency. In previous publications by our group, we showed that different CBD-rich Cannabis extracts, with equal amounts of CBD but varying concentrations of other minor compounds, lead to diverse anticonvulsant effects in a mouse model of epilepsy (29), and selective antitumoral effects in Notch1-mutated in T-ALL (32); suggesting the potential therapeutic effects of other compounds in the extracts in addition to CBD.

A possible mechanism previously suggested to explain the difference between the effects of purified phytocannabinoids versus full-spectrum extracts is the "entourage effect", where one compound may synergize the activity and efficacy of another on the same target (48, 49). This "entourage effect" was postulated by Russo (48) based on a phenomenon observed in the eCBS by Ben Shabbat *et al.* (51). While it is well-established for endocannabinoids, only very few studies demonstrated this for phytocannabinoids. We showed here that the presence of small amounts of the minor cannabinoids 331-18A and CBDV enhanced the effect of CBD by itself. Our success in isolating and characterizing a minimal number of cannabinoids that achieve the same antitumor effect as the whole extract is a significant advancement in the field of medical Cannabis and provides solid evidence for the "entourage effect".

Subsequent reports have expanded the notion of the "entourage effect" to polypharmaceutical and/or polypharmacological effects, where several of the plant metabolites work simultaneously on the same receptors, and/or when the same molecule has several different targets within the same system (52–54). CBD has been shown to present complex signaling capabilities of many receptors in the extended eCBS. Specifically, it was shown to have a weak affinity to CB2, and a strong affinity to many of the TRPV family receptors, including TRPV1, TRPV2 and TRPV3 (55, 56).

We suggest that the Notch1-mutated T-ALL cells are affected through the engagement of the eCBS receptor CB2 and the channel TRPV1. More specifically, the known cannabinoids CBD and CBDV affected both, while the novel 331-18A exerted its effect only through CB2. Following the stimulation of the receptors,  $\text{Ca}^{2+}$  is depleted from the ER and there is an increase in cytosolic  $\text{Ca}^{2+}$  and activation of eIF2 $\alpha$  followed by ATF4-CHOP-CHAC1 signaling. Notably, the inhibition of eIF2 $\alpha$  upstream to ATF4 rescued T-ALL cells from reduced NICD expression by treatment with 331-18A, CBDV and CBD combination. The upregulation of CHAC1 inhibits the S1 furin-like cleavage of Notch1 at the Golgi apparatus (35), leading to the arrival of immature Notch1 to the plasma membrane as we have previously shown (32). This inhibition of the Notch1 pathway ultimately leads to apoptosis in T-ALL cells that rely on mutated Notch1 for propagation (see schematic representation in Fig. 7).





**Fig. 7.**

### Schematic representation.

(A) In WT cells, the Notch1 receptor is only activated upon ligand binding. In T-ALL cells that have a Notch1 mutation, the receptor is constitutively active regardless of ligand binding, leading to a positive feedback loop in which the mature receptor is cleaved by a series of sequential cleavage events, causing the release of NICD, which translocates to the nucleus where it promotes transcription of target genes involved in cell growth. The premature Notch1 is cleaved by a furin-like convertase in the trans-Golgi apparatus, resulting in a mature receptor that is transported to the membrane. (B) The three phytocannabinoids CBD, CBDV and 331-18A in the whole extract stimulate CB2 and TRPV1 leading to an increase in cytosolic  $Ca^{2+}$  and ER stress induced activation of eIF2 $\alpha$  followed by ATF4-CHOP-CHAC1 signaling. CHAC1 inhibits the furin-like cleavage of Notch1, preventing its maturation and resulting in reduced NICD formation, reduced cell viability and increased apoptosis.

The potential antitumor effects of cannabinoids through the Notch1 signaling pathway were previously demonstrated in glioma, where THC induced the expression of p8 protein, upstream to ATF3/4 (57). The activation of Notch signaling by ER stress was demonstrated in additional systems, including Alzheimer's disease (58), polycystic ovary syndrome (59) and other types of cancer (60, 61). A previous work that focused on chronic lymphocytic leukemia suggested a role for ER stress in Notch1 cleavage and leukemia cell death through the inhibition of proteasome activity (62). These findings suggest blocking Notch1 maturation through the eIF2 $\alpha$ -ATF4-CHOP-CHAC1 pathway and inhibiting the S1 furin-like cleavage of Notch1 have the potential to affect all types of activation mutations in Notch1, and also all four human homologs of Notch (Notch1-4) (35, 63).

We demonstrated the effectiveness of the whole extract and the combination of the cannabinoids in several *in-vivo* models. To validate the clinical relevance of our findings for patients, we tested the extract containing the three cannabinoids in an *in-vivo* PDX model (64, 65) (fig. S7), and demonstrated treatment reduced leukemic burden. As medical Cannabis is already routinely prescribed to cancer patients, tailoring chemovars that contain the appropriate cannabinoids to T-ALL patients can provide them with optimal care that offers anti-cancer properties on top of the palliative ones. Our findings are potentially relevant for numerous diseases, as dysregulation of Notch signaling has been found in various other cancers including breast, prostate and colorectal; as well as in non-cancerous diseases (8, 9, 12). As such, they may pave the way for the establishment of new pharmacotherapies for the treatment of Notch1-dependent malignancies and other diseases.

## Materials and Methods

### Reagents

The analytical standards (>98%) for phytocannabinoids were purchased from Silicol Scientific Equipment Ltd. (Or Yehuda, Israel). The antagonists CID 16020046 (#4959/5), AM630 (#1120/10), SR-144,528 (#5039/10) and AMG9810 (#2316/10) were purchased from Tocris Bioscience (Bristol, UK). BIM-46187 (#5332990001) and ISRIB (#SML0843) were purchased from Sigma-Aldrich (Rehovot, Israel). Synthetic CBD, CBDV and 331-18A were purchased from Cannasoul Analytics (Caesarea, Israel).

### Phytocannabinoids analysis

Comprehensive phytocannabinoid analysis was performed using a UHPLC system coupled to a Q Exactive™ Focus Hybrid Quadrupole-Orbitrap MS (Thermo Scientific, Bremen, Germany) using a similar chromatographic method as previously described (29, 32). MS acquisition was carried out with a heated electro spray ionization (HESI-II) ion source operated in negative mode. Source parameters were as follows: sheath gas flow rate, auxiliary gas flow rate and sweep gas flow rate: 50, 20 and 0 arbitrary units respectively; capillary temperature: 350°C; heater temperature: 50°C; spray voltage: 3.00 kV. The scan range was 150–550 m/z for all acquisition events. MS was operated in full MS<sup>1</sup> mode at 70,000 resolution, and the AGC target was set to 10<sup>6</sup> with a maximum IT of 100 ms. Identification and absolute quantification of phytocannabinoids was performed by external calibrations as described by Berman *et al.* (29). Absolute quantification of phytocannabinoids with analytical standards was performed by external calibrations. Standard mixes were prepared ranging from 1 to 1000 ng/mL for  $\Delta^9$ -THCA,  $\Delta^9$ -THC, CBDA and CBD, and 0.25 to 625 ng/mL for all the other components. The dynamic range for each component was determined as maximum deviation from expected concentrations of 20%, and minimum signal-to-noise ratios of 10. All other phytocannabinoids without analytical standards were semi-quantified according to calibration curves from the same phytocannabinoid family or average curves for unknown compounds (29). All samples were injected and analyzed by ESI-LC/MS in three concentrations (10, 1 and 0.1  $\mu$ g/mL Cannabis extract to ethanol).

## Isolation of single phytocannabinoids

Fractionation of the whole Cannabis extract into four fractions and isolation of single phytocannabinoids were performed using a semi-preparative HPLC/UV (Thermo Scientific, Bremen, Germany). The chromatographic separation was achieved using a Luna® C18 column (10 µm, 250 mm × 21.1 mm i.d.) and a two-solvent A/B multistep gradient (solvent A: 0.1% acetic acid in water and solvent B: 0.1% acetic acid in acetonitrile, the solvents and acetic acid were of HPLC and LC/MS grades, respectively). The multistep gradient program was established as follows: initial conditions were 50% B for 4 min, raised to 76% B until 6.5 min, held at 76% B for 16.5 min, and then raised to 90% B until 26 min, held at 90% B until 31 min, decreased to 50% B over the next 4 min, and held at 50% B until 40 min for re-equilibration of the system prior to the next injection. A flow rate of 20 ml/min and an injection volume of 500-750 µl were used. Data acquisition was performed at 220 nm. The crude Cannabis extract was prepared in a concentration of 100 mg/ml in ethanol. Fractions were collected in 10 min intervals and then lyophilized to dryness and analyzed by UHPLC/UV and ESI-LC/MS.

To improve the yield and purity of the fractions for isolation of single phytocannabinoids, a preliminary separation step was employed prior to semi-preparative HPLC fractionation. The crude Cannabis extract was dissolved to a concentration of 0.5 g/ml in ethanol, filtered using a PTFE 0.45 µm filter, and then purified using a PLC 2050 Purification System, equipped with a CPC-250 column and a photodiode array (PDA) detector (Gilson Inc.). Separations were achieved using a biphasic solvent system consisting of upper (hexane:acetonitrile in the ratio of 98:2 v/v) and mobile (acetonitrile:water in the ratio 50:50 v/v) phases in descending mode. The column was operated with a flow rate of 12.5 ml/min and a rotation speed of 2,200 rpm. The gradient program was established as follows: initial conditions were acetonitrile:water:hexane in the ratio 50:50:0 v/v for 25 min, and then gradually raised to acetonitrile:water:hexane in the ratio 64:34:2 v/v until 45 min. All the solvents were of HPLC grade. The collected fractions were lyophilized to dryness and then subjected to semi-preparative HPLC fractionation. The specific phytocannabinoids were collected according to retention time.

## Structure elucidation

The structure of the isolated 331-18A was elucidated by <sup>1</sup>H and <sup>13</sup>C one-dimensional NMR spectra (Avance-III-700, Bruker, Germany) with CBD as the reference. The samples were dissolved in CDCl<sub>3</sub> and analyzed at 4°C with Tetramethylsilane as the internal reference. The optical rotation of 331-18A in comparison to CBD was determined in chloroform (10 mg/mL) using a Jasco P-1010 (Rudolph Research Analytical, US) polarimeter at 26°C.

## Cell culture

The T-cell leukemia cell lines MOLT-4 (CVCL\_0013, CRL-1582™) and CCRF-CEM (CVCL\_0207, CCL-119™) were purchased from ATCC. Cell lines were grown in RPMI 1640 media (Sigma-Aldrich; R8758), and kept at 37°C in a 5% carbon dioxide incubator. Unless otherwise stated, all culture media were supplemented with 10% FBS (Biological Industries; 04-007-1A), penicillin (100 U/mL) and streptomycin (100 µg/mL) (Biological Industries; 03-031-1B).

## Cell apoptosis assay

Cells were cultured in 12-well plates at 1×10<sup>6</sup> cells/well with media containing 0.5% FBS and incubated for 24 h with either DMSO (vehicle), Cannabis extract 12, fractions 1-4 and different combinations of 0.06 µg/ml 331-18A, 0.06 µg/ml CBDV and 1.5 µg/ml CBD. Apoptotic cells were detected by an Annexin V/PI assay using flow cytometry or by detection of cleaved caspase-3.

## Annexin V/PI assay

Apoptosis was assessed by Annexin V-FITC (BioVision; 1006-200; 1:500) and PI staining in an annexin binding buffer (BioVision; 1006-100; 1:500) according to the manufacturer's instructions. Apoptosis assessed by flow cytometry used a BD™ LSR II digital four-laser flow cytometer (BD Biosciences) and was analyzed by BD FACSDiva™ software version 6.1.2. (BD Biosciences). Results were calculated as percentage of positive Annexin V-FITC cells out of total cells counted (30,000 events).

## Western blot analysis

Cells were quickly washed twice with ice-cold PBS, and lysed in radioimmunoprecipitation assay buffer (RIPA) lysis buffer (Sigma Aldrich; R0278) freshly supplemented with protease and phosphate inhibitors (Cell Signaling Technology; 5872). Lysates were cleared by centrifugation at 13,000 g at 4°C for 10 min. Protein concentration was determined by a Bradford assay (Sigma Aldrich; B6916). Proteins were denatured in NuPAGE™ LDS Sample Buffer (4X) (Thermo Fisher; NP0008) and supplemented with β-mercaptoethanol (Sigma Aldrich; B6131). Typically, 20-50 mg of total protein were loaded per lane on a Novex™ 4-20% Tris-Glycine gradient gel (Invitrogen; XP04202BOX) and electrophoretically transferred to a nitrocellulose membrane (Bio-Rad; 1704159S). Membranes were blocked with TBS 0.1% Tween 20 buffer containing 5% bovine serum albumin (Sigma; A7906) for 1 h and incubated with the primary antibody overnight at 4°C. This was followed by incubation with horseradish peroxidase-labeled matching secondary antibodies. Immunoreactive bands were detected by Luminata™ HRP substrate (Millipore; WBLUR0500) and visualized using a MicroChemi imager (DNR Bioimaging Systems). Antibodies used: Cleaved Notch1 Val1744 (CST, 4147S; AB\_2153348; 1:1,000) to assess NICD, cleaved caspase-3 (CST, 9664; AB\_2070042; 1:1,000), β-tubulin (CST; clone D3U1W, 86298; AB\_2715541; 1:1,000), GAPDH (CST, 2118; AB\_561053; 1:5,000), ATF4 (Proteintech, 10835-1-AP; AB\_2058600; 1:500), CHOP (Proteintech, 15204-1-AP; AB\_2292610; 1:500), CHAC1 (abcam, ab76386; AB\_1658635; 1:500), eIF2α (CST, 5324; AB\_10692650; 1:1000) and Phospho-eIF2α (Ser51) (CST, 3398; AB\_2096481; 1:1000). Secondary antibodies include: Peroxidase AffiniPure Goat Anti-Rabbit IgG (H+L) (Jackson ImmunoResearch, 111-035-144; AB\_2307391; 1:5,000) and Peroxidase-AffiniPure Goat anti mouse IgG (HZ+L) (Jackson ImmunoResearch, 115-035-003; AB\_10015289; 1:5000).

## RNA extraction

Total RNA was isolated from MOLT-4 cells ( $1 \times 10^6$  cells/sample) using TRIzol® (Thermo Fisher; 15596026) and RNeasy kit (Qiagen; 74104) according to the manufacturer's instructions. Sample quality was assessed both spectrophotometrically (Nanodrop Technologies®) and by gel electrophoresis with 1% agarose.

## RNA-seq

RNA sequencing of differentially expressed genes was performed on extracted RNA with an Affymetrix Clariom-S microarray for human genome RNA expression analysis and an Affymetrix Transcriptome Analysis Console software. The threshold set for increased- and decreased-abundance genes was a fold change  $\geq 1.5$  and  $p < 0.05$ . The sequencing was performed by the Center for Genomic Technologies of The Hebrew University of Jerusalem.

## Quantitative RT-PCR

cDNA was synthesized from 1 µg of extracted RNA. Purified RNA was reverse transcribed with the qScript™ cDNA synthesis kit (Quanta Biosciences; 95047). The mRNA expression levels of were quantified using qRT-PCR 7300 system (Applied Biosystems) using Human Taqman probes (Thermo Fisher Scientific) for: *GAPDH* (Hs02758991\_g1); *CNR1* (CB1) (Hs01038522\_s1); *CNR2* (CB2) (Hs00361490\_m1); *GPR55* (Hs00271662\_s1); *TRPV1* (Hs00175798\_m1); *TRPM8* (Hs01066596\_m1);

*TRPA1* (Hs00175798\_m1); *ATF4* (Hs00909569\_g1); *DDIT3* (CHOP) (Hs00358796\_g1); *CHAC1* (Hs00225520\_m1). Relative expression values were normalized using to the endogenous housekeeping gene *GAPDH* as the control and calculated using standard  $\Delta$ CT methods.

## Calcium imaging

MOLT-4 Cells were loaded with the fluorescein  $\text{Ca}^{2+}$  probe Fluo-4 AM (Invitrogen # F14217). Cells were incubated for 60 min in RPMI medium and supplemented with 1  $\mu\text{M}$  Fluo-4/AM. Cells were then rinsed twice and incubated for another 30 min in a Fluo-4 AM-free HBSS buffer (Sigma Aldrich #H6648). During these steps the cells were kept in the dark. Fluo-4 signals were collected by a BD™ LSR II digital four-laser flow cytometer (BD Biosciences) and analyzed by BD FACSDiva™ software version 6.1.2. (BD Biosciences).

## Cell viability assay

Cells were cultured in 96-well plates. At  $1 \times 10^5$  cells/well the media was replaced with RPMI containing 0.5% FBS and treatments as indicated. DMSO was used as the vehicle. The XTT-based cell proliferation kit (Sartorius, Beit Haemek, Israel) was used to quantitate viability according to the manufacturer's instructions. Cytotoxicity was measured following 24 hrs of incubation using fluorescence spectrophotometry at excitation and emission wavelengths of 450-500 and 630-690 nm, respectively, and calculated as percent reduction between treated and control cells.

## Synergy Scoring

The cell viability percentage for each concentration was assessed with an XTT-based cell proliferation kit (Sartorius, Beit Haemek, Israel), then entered into SynergyFinder web application (v 2.0). The HSA model was used to calculate synergy scores of the cannabinoid combinations. Estimation of outlier measurements was carried out with cNMF algorithm ([66](#)) implemented in SynergyFinder.

## Mice and procedures

All procedures were performed according to protocols approved by the Technion Administrative Panel of Laboratory Animal Care (animal protocol number: IL\_0470317). Female NOD.CB17-Prkdcscid/NCrHsd (IMSR\_ENV:HSD-170) adult (6-7 weeks) mice were purchased from Envigo. Female NSG adult (6-7 weeks) mice were purchased from the Technion animal facility.

For the in-vivo tumorigenicity assay, MOLT-4 cells mixed with BD Matrigel™ Matrix Growth Factor Reduced (BD; FAL356230) (1:1 v/v) were injected subcutaneously into the flanks of 7-week-old female NOD-scid mice. Tumor size was measured with a caliper, and the tumor volume was determined according to the formula: length  $\times$  width<sup>2</sup>  $\times$  0.5. The mice were euthanized after the indicated days as none reached the maximal endpoint size (1  $\text{cm}^3$ ). The xenograft tumors were dissected and weighed.

For the in-vivo leukemogenesis assay and survival analysis, CCRF-CEM cells were intravenously injected to 7-week female NSG mice. The bone marrow was harvested from each mouse and the infiltrated CCRF cells were identified by staining for a human CD45 antibody (Biolegend; AB\_2687375; BLG-368522; 1:200).

For the Patient-Derived Xenograft (PDX) model, frozen aliquots of T-ALL patient samples in which Notch1 mutation was verified by next-generation sequencing were purchased from Dana-Farber Cancer Institute. The cells were expanded in mice and injected subcutaneously into the flanks of 7-week female NSG mice. To investigate the inhibitory effects of extract 12 on leukemogenesis, 30 mg/kg/day extract was intraperitoneally injected three times a week, starting three weeks after cell

injection. The mice were euthanized after the indicated days. BM, peripheral blood and spleens were harvested from each mouse and the infiltrated primary cells were identified using flow cytometry by staining for human CD45 antibody (BioLegend; BLG-368522; 1:200).

## Cannabinoids analysis from tumor tissue

Fresh excised tumors were accurately weighed and dissected to smaller sections. Hanks' Balanced Salt Solution (Sigma Aldrich; H8264) was added in a ratio of 1 ml per 100 mg sample. The samples were further homogenized into single-cell suspensions with a gentleMACS™ Dissociator (Miltenyi Biotec) using a pre-set program for tumor tissues. The single cell suspensions were stored at -80°C until further extraction. CBD, CBDV and 331-18A were analyzed using the ESI-LC/MS system by retention time, accurate mass and spectral matching against our developed ESI-LC/MS/MS library of phytocannabinoids (29). Absolute quantification of the three compounds was performed by the stable isotope dilution method as recently described by our group (40). Briefly, the extraction solution (methanol:acetonitrile:acetic acid in the ratio 50:50:0.1 v/v) spiked with 20 ng/mL of CBD-d3 internal standard (Sigma-Aldrich; C-084) was added to the single cell suspensions in the ratio 3:1 v/v, respectively. Samples were thoroughly vortexed and then centrifuged for 20 min at 4°C for protein and cell precipitation. The supernatants were then mixed in a ratio of 1:3 v/v sample to 0.1% v/v acetic acid in water, and loaded onto Agela Cleanert C8 solid phase extraction cartridges (500 mg of sorbent, 50 µm particle size). Cannabinoids were eluted from the columns using 2 ml of 0.1% v/v acetic acid in methanol, evaporated to dryness by speedvac, and reconstituted in 100 µl ethanol. Since there was no analytical standard available for 331-18A, this compound was semi-quantified according to the calibration curve of CBD.

## Statistics and reproducibility

Data were analyzed with GraphPad Prism software version 7.04. Statistical significance was determined by either an unpaired two-tailed Student's *t*-test (with unequal variations if an F test ruled out the equal variation assumption) or one-way ANOVA followed by Sidak's multiple comparisons test for multiple groups. Kaplan–Meier survival curves were analyzed with the Gehan–Breslow–Wilcoxon test, as appropriate for the dataset. Data are generally expressed as mean ± SEM, as denoted in the Figure captions, with  $p \leq 0.05$  was considered statistically significant.

## Supplementary Materials

**Table S1.** Phytocannabinoid concentrations by UHPLC/LC-MS of fractions relative to the whole extract.

**Table S2.** Phytocannabinoid concentrations by UHPLC/LC-MS of cannabis fraction 2 peaks.

**Table S3.**  $^1\text{H}$  and  $^{13}\text{C}$  peak assignments and chemical shifts of 331-18A.

**Table S4.** Ten most increased- and decreased-abundance genes following treatment of MOLT-4 cells with the whole extract according to Affymetrix.

**Fig. S1.** Fraction 2 is as effective as the whole extract in reducing the viability of MOLT-4 cells.

**Fig. S2.** Specific identification of phytocannabinoids with anti-cancer properties by spectral matching and peak purity.

**Fig. S3.** CBD and 331-18A NMR spectra.

**Fig. S4.** Notch1 downregulation by Extract 12 is mediated through CB2 and TRPV1 followed by ATF4-CHOP-CHAC1 signaling pathway.



**Fig. S5.** Whole extract treatment activates eIF2 $\alpha$ .

**Fig. S6.** Inhibition of tumor growth *in-vivo* by whole extract.

**Fig. S7.** Treatment with extract containing the three cannabinoids inhibits leukemic expansion in an *in-vivo* PDX model.

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## Author contributions

Conceptualization: E.B., S.P. and D.M.

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Analysis and interpretation of data: E.B., A.G., P.B. and S.P.

Writing-original draft: E.B. and D.M.

Writing-review & editing: E.B., D.M., P.B. and S.P.

Study supervision: D.M.

## Competing interests

A patent pertaining to the results of the paper has been applied for the use of cannabinoids in the treatment of Notch-related diseases. This patent was licensed to Cavnox Ltd. to test the combination of the molecules in human clinical trials.

## Data and materials availability

The RNA-Seq data that support the findings of this study have been deposited in the Gene Expression Omnibus under the accession code GSE154287. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE154287> .

All data generated or analyzed during this study are included in this published article and its supplementary information files.

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## Editors

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**Reviewer #2 (Public Review):**

**Summary:**

The Meiri group previously showed that Notch1-activated human T-ALL cell lines are sensitive to a cannabis extract in vitro and in vivo (Ref. 32). In that article, the authors showed that Extract #12 reduced NICD expression and viability, which was partially rescued by restoring NICD expression. Here, the authors have identified three compounds of Extract #12 (CBD, 331-18A, and CBDV) that are responsible for the majority of anti-leukemic activity and NICD reduction. Using a pharmacological approach, the authors determined that Extract #12 exerted its anti-leukemic and NICD-reducing effects through the CB2 and TRPV1 receptors. To determine mechanism, the authors performed RNA-seq and observed that Extract #12 induces ER calcium depletion and stress-associated signals – ATF4, CHOP, and CHAC1. Since CHAC1 was previously shown to be a Notch inhibitor in neural cells, the authors assume that the cannabis compounds repress Notch S1 cleavage through CHAC1 induction. The induction of stress-associated signals, Notch repression, and anti-leukemic effects were reversed by the integrated stress response (ISR) inhibitor ISRIB. Interestingly, combining the 3 cannabinoids gave synergistic anti-leukemic effects in vitro and had growth inhibitory effects in vivo.

**Strengths:**

- (1) The authors show novel mechanistic insights that cannabinoids induce ER calcium release and that the subsequent integrated stress response represses activated NOTCH1 expression and kills T-ALL cells.
- (2) This report adds to the evidence that phytocannabinoids can show a so-called "entourage effect" in which minor cannabinoids enhance the effect of the major cannabinoid CBD.
- (3) This report dissects out the main cannabinoids in the previously described Extract #12 that contribute to T-ALL killing.
- (4) The manuscript is clear and generally well-written.
- (5) The data are mostly high quality and with adequate statistical analyses.
- (6) The data generally support the authors' conclusions. The main exception is the experiments related to Notch.
- (7) The authors' discovery of the role of the integrated stress response might explain previous observations that SERCA inhibitors block Notch S1 cleavage and activation in T-ALL (Roti Cancer Cell 2013). The previous explanation by Roti et al was that calcium depletion causes Notch misfolding, which leads to impaired trafficking and cleavage. Perhaps this explanation is not entirely sufficient?

**Weaknesses:**

- (1) Given the authors' previous Cancer Communications paper on the anti-leukemic effects and mechanism of Extract #12, the significance of the original manuscript was reduced. To increase significance, the authors provided a new Fig. S7 in the revision showing that Extract #12 inhibits PDX growth in vivo. This experiment is nicely supportive of the anti-leukemic effects of Extract #12, raising the significance of their previous Cancer Communication paper by using in vivo patient-derived cells. However, this reviewer had suggested testing the

combination of 333-18A+CBVD+CBD since the combination is the focus of the current manuscript. For unclear reasons, the combination was not tested.

(2) It would be important to connect the authors' findings and a wealth of literature on the role of ER calcium/stress on Notch cleavage, folding, trafficking, and activation. The several references suggested by this reviewer were not included in the revised manuscript for unclear reasons. These references are important to show the current status of the field and help readers appreciate what this manuscript brings that is new to T-ALL. In particular, Roti et al (Cancer Cell 2013) showed that SERCA inhibitors like thapsigardin reduce ER calcium levels and block Notch signaling by inhibiting NOTCH1 trafficking and inhibiting Furin-mediated (S1) cleavage of Notch1 in T-ALL. Multiple EGF repeats and all three Lin12/Notch repeats in the extracellular domains of Notch receptors require calcium for proper folding (Aster Biochemistry 1999; Gordon Nat. Struct. Mol. Biol. 2007; Hambleton Structure 2004; Rand Protein Sci 1997). Thus, Roti et al concluded that ER calcium depletion blocks NOTCH1 S1 cleavage in T-ALL. This effect seems to be conserved in Drosophila as Periz and Fortiini (EMBO J, 1999) showed impaired Notch cleavage in  $Ca^{2+}$ /ATPase-mutated Drosophila cells.

(3) There is an overreliance of the data on single cell line – MOLT4. MOLT4 is a good initial choice as it is Notch-mutated, Notch-dependent, and representative of the most common T-ALL subtype – TAL1. However, there is no confirmatory data in other TAL1-positive T-ALLs or interrogation of other T-ALL subtypes. While this reviewer appreciates that the authors showed that multiple T-ALL cell lines were killed in response to Extract #12 in a previous study, the current manuscript is a separate study that should stand on its own. T-ALLs can be killed by multiple mechanisms. It would be important to show a few pieces of key data illustrating that the mechanism of killing found in MOLT4 applies to other T-ALLs.

(4) Fig. 6H. The effects of the cannabinoid combination might be statistically significant but seems biologically weak.

(5) Fig. 3. Based on these data, the authors conclude that the cannabinoid combination induces CHAC1, which represses Notch S1 cleavage in T-ALL cells. The concern is that Notch signaling is highly context dependent. CHAC1 might inhibit Notch in neural cells (Refs. 34-35), but it might not do this in a different context like T-ALL. It would be important to show evidence that CHAC1 represses S1 cleavage in the T-ALL context. More importantly, Fig. 3H clearly shows the cannabinoid combination inducing ATF4 and CHOP protein expression, but the effects on CHAC1 protein do not seem to be satisfactory as a mechanism for Notch inhibition. Perhaps something else is blocking Notch expression?

In the rebuttal, the two references provided by the authors do not alleviate concern that CHAC1 might not be acting as a Notch cleavage inhibitor in the T-ALL context. The Meng et al paper studied B-ALL not T-ALL and did not evaluate CHAC1 as a possible Notch cleavage inhibitor. Likewise, the Chang et al paper did not evaluate CHAC1 as a possible Notch cleavage inhibitor. Therefore, whether CHAC1 is a Notch cleavage inhibitor in the T-ALL context remains an open question. While the authors are correct that Supplementary Fig. S4G-I show that Extract #12 clearly induces CHAC1 protein expression, Main Fig. 3H shows that the extract combination 333-18A+CBVD+CBD, which is the focus of this manuscript, has unclear effects. If the extract combination has no effect on CHAC1 but has the same effects on Notch1 expression as the full extract, then there is reduced support for the authors' conclusion that the full extract and the 333-18A+CBVD+CBD combination inhibit Notch through CHAC1 induction.

(6) The authors provide a new figure on page 5 of the rebuttal that was not requested. It is supposed to show that CHAC1 loss protects T-ALL cells from Extract #12-induced cell population decline. Unfortunately, this figure is not conclusive. The empty vector PLKO is not an appropriate negative control. The field uses non-targeting shRNA controls like pLKO-luciferase to control for induction of the RNA interference pathway. Further, the viability

data in panel B is normalized such that the effect of shCHAC1 on viability is masked. Showing non-normalized data is important, because if shCHAC1 impairs viability compared to control shRNA, then CHAC1 might have effects on non-Notch pathways, which would reinforce the above concern in Point #5 that CHAC1 might not act as a Notch inhibitor in the T-ALL context. Separately, if this experiment had tested whether CHAC1 knockdown increases Notch cleavage and Notch target gene expression like DTX1, HES1 and MYC, then such data would have helped address Point #5.

(7) Fig. 4B-C/S5D-E. These Western blots of NICD expression are consistent with the cannabinoid combination blocking Furin-mediated NOTCH1 cleavage, which is reversed by ISR inhibition. However, there are many mechanisms that regulate NICD expression. To support their conclusion that the effects are specifically Furin-mediated, the authors should probe full length (uncleaved) NOTCH1 in their Western blots. While the authors showed that the full extract (#12) increased uncleaved NOTCH1 expression in their Cancer Communications paper, a major conclusion of the manuscript is that the cannabinoid combination 333-18A+CBVD+CBD reproduces the effect of the full extract (#12). To support this conclusion, the authors should probe key blots for full-length Notch to show that the cannabinoid combination increases uncleaved NOTCH1 just like Extract #12 did in the authors' Cancer Communications paper.

(8) Fig. S4A-B. While these pharmacologic data are suggestive that Extract #12 reduces NICD expression through the CB2 receptor and TRPV1 channel, the doses used are very high (50uM). To exclude off-target effects, these data should be paired with genetic data to support the authors' conclusions. In the rebuttal, the authors provide dose response cell viability curves of the CB2 and TRPV1 inhibitors. These curves do not exclude the possibility that 50uM has off-target effects. This reviewer notes that Reviewer #1 had similar concerns and that both reviewers requested genetic validation of the pharmacological data. These data were not provided in the revision.

(9) Since the authors have performed gene expression profiling, an orthogonal test to confirm that Extract #12 acts through the Notch pathway is to perform enrichment analysis using Notch target gene signatures in T-ALL (e.g. Wang PNAS 2013). In contrast to the authors' rebuttal, this reviewer does not see any enrichment analysis (e.g. GSEA plots) performed on the microarray data to show that Extract #12 inhibits the Notch pathway.

(10) The revised manuscript still retains references that microarray data are "RNA-seq" data, which is inaccurate (see page 10, line 160; Figure 3 legend; page 12, line 169; page 27, line 428; page 36, line 741)

<https://doi.org/10.7554/eLife.90854.2.sa0>

## Author Response

The following is the authors' response to the original reviews.

### **Reviewer #1 (Public Review):**

*This is an interesting manuscript that extends prior work from this group identifying that a chemovar of Cannabis induces apoptosis of T-ALL cells by preventing NOTCH1 cleavage. Here the authors isolate specific components of the chemovar responsible for this effect to CBD and CBDV. They identify the mechanism of action of these agents as occurring via the integrated stress response. Overall the work is well performed but there are two lingering questions that would be helpful to address as follows:*

- *Exactly how CBD and CBDV result in the upregulation of the TRPV1/integrated stress response is unclear. What is the most proximal target of these agents that results in these changes?*

The interaction of CBD and CBDV with TRPV1 has been thoroughly investigated by previous studies in the field. A few prominent examples are:

- (1) De Petrocellis, Luciano, Alessia Ligresti, Aniello Schiano Moriello, Marco Allarà, Tiziana Bisogno, Stefania Petrosino, Colin G. Stott, and Vincenzo Di Marzo. "Effects of cannabinoids and cannabinoid-enriched Cannabis extracts on TRP channels and endocannabinoid metabolic enzymes." *British journal of pharmacology* 163, no. 7 (2011): 1479-1494.
- (2) Muller, Chanté, Paula Morales, and Patricia H. Reggio. "Cannabinoid ligands targeting TRP channels." *Frontiers in molecular neuroscience* 11 (2019): 487.
- (3) Iannotti, Fabio Arturo, Charlotte L. Hill, Antonio Leo, Ahlam Alhusaini, Camille Soubrane, Enrico Mazzarella, Emilio Russo, Benjamin J. Whalley, Vincenzo Di Marzo, and Gary J. Stephens. "Nonpsychotropic plant cannabinoids, cannabidivarin (CBDV) and cannabidiol (CBD), activate and desensitize transient receptor potential vanilloid 1 (TRPV1) channels in vitro: potential for the treatment of neuronal hyperexcitability." *ACS chemical neuroscience* 5, no. 11 (2014): 1131-1141.
- (4) Costa, Barbara, Gabriella Giagnoni, Chiara Franke, Anna Elisa Trovato, and Mariapia Colleoni. "Vanilloid TRPV1 receptor mediates the antihyperalgesic effect of the nonpsychoactive cannabinoid, cannabidiol, in a rat model of acute inflammation." *British journal of pharmacology* 143, no. 2 (2004): 247-250.
- (5) de Almeida, Douglas L., and Lakshmi A. Devi. "Diversity of molecular targets and signaling pathways for CBD." *Pharmacology research & perspectives* 8, no. 6 (2020): e00682.
- (6) Anand, Uma, Ben Jones, Yuri Korchev, Stephen R. Bloom, Barbara Pacchetti, Praveen Anand, and Mikael Hans Sodergren. "CBD effects on TRPV1 signaling pathways in cultured DRG neurons." *Journal of Pain Research* (2020): 22692278.

Similarly, other works have demonstrated the link between TRPV1 and the integrated stress response pathway (see below). These studies suggested increased reactive oxygen species (ROS) production, Cyclooxygenase-2 (COX-2) enzyme, as well as other stressors, lead to modulation of intracellular calcium levels by TRPV1.

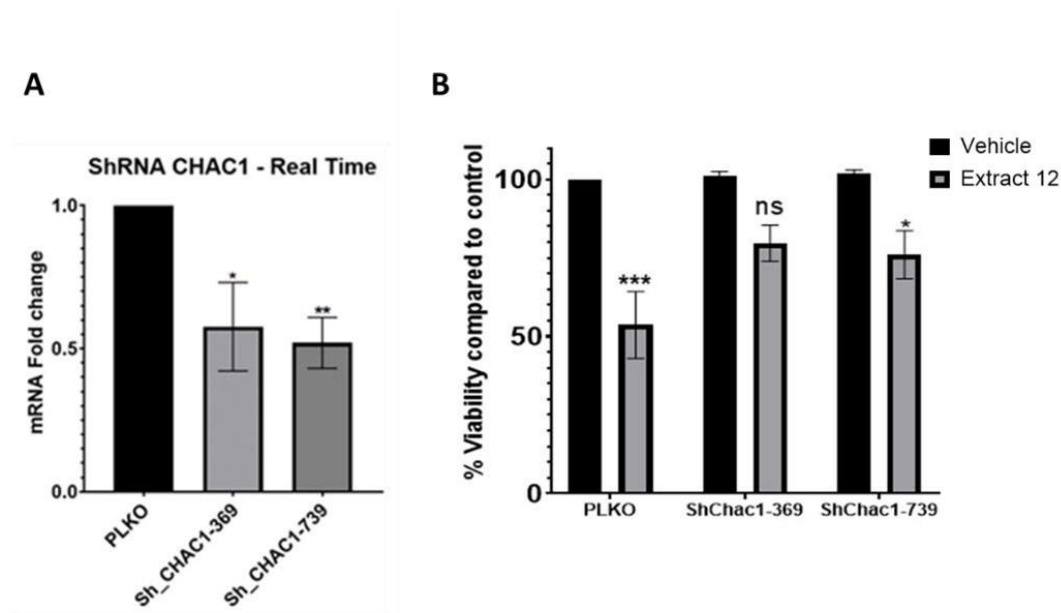
- (1) Ho, Karen W., Nicholas J. Ward, and David J. Calkins. "TRPV1: a stress response protein in the central nervous system." *American journal of neurodegenerative disease* 1, no. 1 (2012): 1.
- (2) de la Harpe, Amy, Natasha Beukes, and Carminita L. Frost. "CBD activation of TRPV1 induces oxidative signaling and subsequent ER stress in breast cancer cell lines." *Biotechnology and Applied Biochemistry* 69, no. 2 (2022): 420-430.
- (3) Soliman, Eman, and Rukiyah Van Dross. "Anandamide-induced endoplasmic reticulum stress and apoptosis are mediated by oxidative stress in nonmelanoma skin cancer: Receptor-independent endocannabinoid signaling." *Molecular Carcinogenesis* 55, no. 11 (2016): 1807-1821.

- *Related to the above, all experiments to confirm the mechanism of action of CBD/CBDV rely on chemical agents, whose precise targets are not fully clear in some cases. Thus, some use of genetic means (such as by knockout of TRPV1, ATF4) to confirm the dependency of these pathways on drug response and NOTCH cleavage would be very helpful.*

Knockdown experiments and inhibition with inhibitors are two different approaches to studying the function of a specific gene or protein. Each method has its advantages and limitations. We initially attempted to knock-down CHAC1, but only managed to silence ~50% (Incomplete knockdown). Following treatment of MOLT4 cells with the whole extract, we observed only a partial downregulation in the mRNA expression of the Notch intracellular domain (NICD) (left panel), which could account for the incomplete rescue from the extract-induced death (right panel). We therefore turned to confirm the signaling pathway by inhibition of different targets with chemical agents.

# Author response image 1.

Partial knockdown of CHAC1 hinders extract-induced cell death. (A) MOLT-4 cells were treated with either an empty vector or shRNA for Chac1, 369 and 739 represent two different areas of Chac1, for 48 hrs. Then, the gene expression of CHAC1 was assessed via qRT-PCR (N=3). (B) MOLT-4 cells were treated as in A, then added vehicle control or whole Extract (3 µg/mL) for additional 24 hrs, and the viability of the cells was assessed with XTT.



## Reviewer #2 (Public Review):

### Summary:

The Meiri group previously showed that Notch1-activated human T-ALL cell lines are sensitive to a cannabis extract in vitro and in vivo (Ref. 32). In that article, the authors showed that Extract #12 reduced NICD expression and viability, which was partially rescued by restoring NICD expression. Here, the authors have identified three compounds of Extract #12 (CBD, 331-18A, and CBDV) that are responsible for the majority of anti-leukemic activity and NICD reduction. Using a pharmacological approach, the authors determined that Extract #12 exerted its anti-leukemic and NICD-reducing effects through the CB2 and TRPV1 receptors. To determine the mechanism, the authors performed RNA-seq and observed that Extract #12 induces ER calcium depletion and stress-associated signals -- ATF4, CHOP, and CHAC1. Since CHAC1 was previously shown to be a Notch inhibitor in neural cells, the authors assume that the cannabis compounds repress Notch S1 cleavage through CHAC1 induction. The induction of stress-



*associated signals, Notch repression, and anti-leukemic effects were reversed by the integrated stress response (ISR) inhibitor ISRIB. Interestingly, combining the 3 cannabinoids gave synergistic anti-leukemic effects in vitro and had growthinhibitory effects in vivo.*

*Strengths:*

*(1) The authors show novel mechanistic insights that cannabinoids induce ER calcium release and that the subsequent integrated stress response represses activated NOTCH1 expression and kills T-ALL cells.*

*(2) This report adds to the evidence that phytocannabinoids can show a so-called "entourage effect" in which minor cannabinoids enhance the effect of the major cannabinoid CBD.*

*(3) This report dissects the main cannabinoids in the previously described Extract #12 that contribute to T-ALL killing.*

*(4) The manuscript is clear and generally well-written.*

*(5) The data are generally high quality and with adequate statistical analyses.*

*(6) The data generally support the authors' conclusions. The exception is the experiments related to Notch.*

*(7) The authors' discovery of the role of the integrated stress response might explain previous observations that SERCA inhibitors block Notch S1 cleavage and activation in T-ALL (Roti Cancer Cell 2013). The previous explanation by Roti et al was that calcium depletion causes Notch misfolding, which leads to impaired trafficking and cleavage. Perhaps this explanation is not entirely sufficient.*

*Weaknesses:*

*(1) Given the authors' previous Cancer Communications paper on the anti-leukemic effects and mechanism of Extract #12, the significance of the current manuscript is reduced.*

Our original manuscript consisted extensive multidisciplinary research, and we were asked to divide the research work into a paper that focuses on the cannabis plant and another paper that addresses finding the specific molecules and their underlying mechanism.

We understand that our publication of the initial observations with the whole extract dampened the overall novelty presented here, but the previous publication lacked the detailed and strong mechanistic work presented here that explains how the cannabis extract exerted its antitumoral effects.

In addition, the finding of the need for 3 phytocannabinoids and the synergy analysis supplies essential support to the 'entourage effect'. This is a phenomenon in which the presence of minor proportions of cannabinoids and other plant components significantly modulate the effects of the main active components of cannabis and thereby produce more potent or more selective effects than the use of one major cannabinoid alone. It was well-demonstrated for endocannabinoids but was only demonstrated in very few studies for phytocannabinoids.

*(2) It would be important to connect the authors' findings and a wealth of literature on the role of ER calcium/stress on Notch cleavage, folding, trafficking, and activation.*

We mentioned three previous papers (ref. 34-36) that guided us in our investigation. Following this reviewer's comment, we added to the discussion a few lines connecting our findings to previous works on ER stress and Notch activation with the appropriate references.

*(3) There is an overreliance on the data on a single cell line -- MOLT4. MOLT4 is a good initial choice as it is Notch-mutated, Notch-dependent, and representative of the most common T-ALL subtype -- TAL1. However, there is no confirmatory data in other TAL1-positive T-ALLs or interrogation of other T-ALL subtypes.*

As mentioned by the reviewer, this study followed a previous publication in which 7 different cell lines were assessed (MOLT-4, CCRF-CEM, Jurkat, Loucy, HPB-ALL, DND-41 and T-ALL1). MOLT-4 cells were used to investigate the mechanism, both MOLT-4 cells and CCRF-CEM cells were utilized to investigate the effect of the cannabinoid combination or the whole extract in vivo.

*(4) Fig. 6H. The effects of the cannabinoid combination might be statistically significant but seem biologically weak.*

Survival rates are presented in Fig. 6H for the combination of the cannabinoids and in Supplementary Fig. S6C for the whole extract. While this mouse model provides valuable insights, the biological significance and the translation of findings to human patients require cautious interpretation.

*(5) Fig. 3. Based on these data, the authors conclude that the cannabinoid combination induces CHAC1, which represses Notch S1 cleavage in T-ALL cells. The concern is that Notch signaling is highly context-dependent. CHAC1 might inhibit Notch in neural cells (Refs. 34-35), but it might not do this in a different context like T-ALL. It would be important to show evidence that CHAC1 represses S1 cleavage in the T-ALL context. More importantly, Fig. 3H clearly shows the cannabinoid combination inducing ATF4 and CHOP protein expression, but the effects on CHAC1 protein do not seem to be satisfactory as a mechanism for Notch inhibition. Perhaps something else is blocking Notch expression?*

We understand the reviewer's concern. Previous works had shown the upregulation of CHAC1 also in the context of Notch signaling in leukemia, particularly recently also for T-ALL:

(1) Meng, X., Matlawska-Wasowska, K., Girodon, F., Mazel, T., Willman, C.L., Atlas, S., Chen, I.M., Harvey, R.C., Hunger, S.P., Ness, S.A. and Winter, S.S., 2011. GSI-I (Z-LLNle-CHO) inhibits  $\gamma$ -secretase and the proteasome to trigger cell death in precursor-B acute lymphoblastic leukemia. *Leukemia*, 25(7), pp.1135-1146.

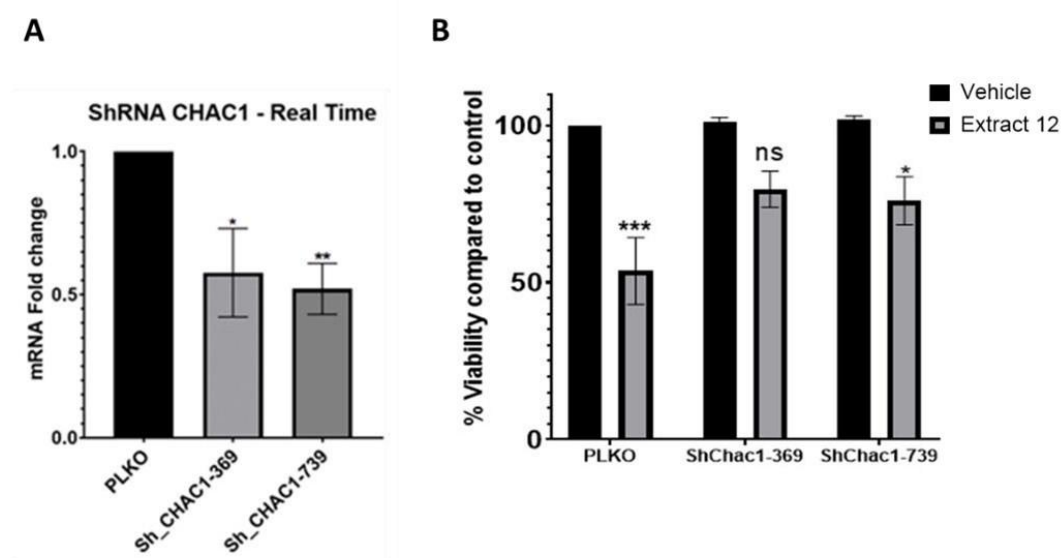
(2) Chang, Yoon Soo, Joell J. Gills, Shigeru Kawabata, Masahiro Onozawa, Giusy Della Gatta, Adolfo A. Ferrando, Peter D. Aplan, and Phillip A. Dennis. "Inhibition of the NOTCH and mTOR pathways by nelfinavir as a novel treatment for T cell acute lymphoblastic leukemia." *International Journal of Oncology* 63, no. 5 (2023): 1-12.

As for the second part of the reviewer's comment, we tested both the mRNA transcript and protein expression of CHAC1. The increase is clearly shown at 60 min for the mRNA Fig. 3D and Fig. 4F and for the protein also in Supplementary Fig. S4G-I.

To show direct involvement of CHAC1 we utilized the means of knockdown. Though it was not completely effective and accounted for about ~50% reduction, it clearly shows the involvement of CHAC1 in the mechanism leading to the reduction in viability of these cancer cells.

## Author response image 2.

Partial knockdown of CHAC1 hinders extract-induced cell death. (A) MOLT-4 cells were treated with either an empty vector or shRNA for Chac1, 369 and 739 represent two different areas of Chac1, for 48 hrs. Then, the gene expression of CHAC1 was assessed via qRT-PCR (N=3). (B) MOLT-4 cells were treated as in A, then added vehicle control or whole Extract (3  $\mu$ g/mL) for additional 24 hrs, and the viability of the cells was assessed with XTT.



(6) Fig. 4B-C/S5D-E. These Western blots of NICD expression are consistent with the cannabinoid combination blocking Furin-mediated NOTCH1 cleavage, which is reversed by ISR inhibition. However, there are many mechanisms that regulate NICD expression. To support their conclusion that the effects are specifically Furin-mediated, the authors should probe full-length (uncleaved) NOTCH1 in their Western blots.

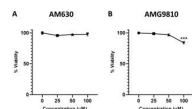
We have probed for the full-length Notch1 in our previously published paper (Cancer Communications, Supplementary Fig. S1G-I). As we have shown here the three cannabinoids together mimic the effect of the whole extract, we did not repeat the experiments with full-length Notch1.

(7) Fig. S4A-B. While these pharmacologic data are suggestive that Extract #12 reduces NICD expression through the CB2 receptor and TRPV1 channel, the doses used are very high (50 $\mu$ M). To exclude off-target effects, these data should be paired with genetic data to support the authors' conclusions.

We performed a dose-response experiment before choosing the doses used for the inhibitors of CB2 and TRPV1 (see below). The dose of 50  $\mu$ M was selected as it did not affect the viability of the cells.

## Author response image 3.

Dose-response of CB2 and TRPV1 inhibitors in MOLT-4 cells. MOLT-4 cells were treated with increasing concentrations ( $\mu$ M) of (A) CB2 inhibitor AM630 or (B) TRPV1 inhibitor AMG9810; and 24 hrs later the viability of the cells was assessed with XTT.



#### Reviewer #2 (Recommendations For The Authors):

(1) In Fig. 6H, it is unclear why the authors are using CCRF-CEM cells, which are known to be resistant to Notch inhibitors, rather than popular cell lines that are Notch-dependent (e.g. CUTLL1, DND-41, HPB-ALL). Since cannabinoids seem to kill at least in part through Notch inhibition, the effects would be predicted to be greater in Notch-dependent T-ALL cell lines than Notch-independent cell lines like CCRF-CEM. To show stronger *in vivo* preclinical efficacy, another suggestion is to combine cannabinoids with tolerable dosing of gammasecretase inhibitors as published by the Michelle Kelliher group.

We have shown in our previous publication that both MOLT-4 and CCRF-CEM cells are dependent on Notch for their propagation, while other cell lines of T-ALL such as Loucy and Jurkat do not. Therefore, we treat CCRF-CEM as Notch-dependent. We discuss the possibility of using the cannabinoid combination with other treatments, specifically chemotherapy, to enhance effectiveness.

(2) To increase significance, this reviewer suggests strengthening the mechanism. However, this reviewer understands the challenge of identifying the correct mechanism. Thus, an alternative would be to increase clinical relevance. Some specific suggestions are described below.

(a) With regard to increasing mechanistic insights, the authors should be aware of some papers that might be helpful. Roti et al (Cancer Cell 2013) showed that SERCA inhibitors like thapsigardin reduce ER calcium levels and block Notch signaling by inhibiting NOTCH1 trafficking and inhibiting Furin-mediated (S1) cleavage of Notch1. Multiple EGF repeats and all three Lin12/Notch repeats in the extracellular domains of Notch receptors require calcium for proper folding (Aster Biochemistry 1999; Gordon Nat. Struct. Mol. Biol. 2007; Hambleton Structure 2004; Rand Protein Sci 1997). Thus, Roti et al concluded that ER calcium depletion blocks NOTCH1 S1 cleavage. This effect seems to be conserved in *Drosophila* as Periz and Fortiin (EMBO J, 1999) showed impaired Notch cleavage in  $Ca^{2+}$ /ATPase-mutated *Drosophila* cells. Besser et al should consider these papers when exploring the mechanism by which the ER calcium release by the cannabinoid combination blocks activated NOTCH1 expression. Similarities and differences should be discussed.

As mentioned above and stated also by the reviewer, many papers have shown the cleavage and/or activation of Notch following ER stress.

(b) With regard to increasing clinical relevance, the authors should consider testing the effects of the cannabinoid combination on primary samples, PDX models, and/or genetically engineered mouse models. Pan-Notch inhibitors like gamma-secretase inhibitors (GSIs) have been disappointing in clinical trials because of excessive on-target toxicity, in particular in the intestine. The authors should consider exploring whether the cannabinoids might be superior to GSIs with regard to intestinal toxicity and why that might be (e.g. receptor expression).

We thank the reviewer and agree that clinical relevance is of utmost importance. As obtaining primary tumor cells from patients is challenging, we assessed the whole cannabis extract in a PDX model. This extract is already being used by patients. We added this result as

Supplementary fig. S7, and address it in the main text of the Results and in the Materials and Methods section.

*(3) Since the authors have performed gene expression profiling, another test to confirm that Extract #12 acts through the Notch pathway is to perform enrichment analysis for known Notch target genes in T-ALL (e.g. Wang PNAS 2013).*

We performed the analysis and this is how we pinpointed the involvement of ATF4, CHOP and CHAC1 of the integrated stress response pathway.

*Minor concern:*

*Supplemental Table S4. According to the text (page 10, line 160) and table title, these data are RNA-seq. However, according to the GSE154287 annotation, these data are Affymetrix arrays There are no gene names in the GSE table. Are the IDs probesets rather than genes?*

Indeed, the gene analysis data are Affymetrix arrays and the title was corrected.